引言

最近在看南京大学李樾和谭添老师的《静态分析》课程，刚好关注了李老师的知乎，看到一个回答是，我们不应该一直沿着别人的思路走，只是纸上谈兵而没有实践。有一位网友提到了Hamming的演讲 You and your research，担心读不下来，就对着翻译软件一边看一遍翻译。结合这两天颜宁教授发表的招生经历，我在开始工作前的前摇太久了，不知道什么是重要的问题，也没有深入地思考过，而且读了本年的论文后发现也没有很好的想法，反而自己的思路似乎被局限住了，偶尔读读其它领域的问题也会感叹原来研究还能这么做。可能这就是一开始走了弯路，走了一年发现自己还在原地踏步。

无论将来是否从事学术界的工作，都应该想办法做点真正的东西出来，别人咀嚼过的东西很多都没办法再去拿来用了。说到底，还是我在最初就给自己设限了，这属实不应该。

希望自己以后也能养成好的习惯，按时做规划和总结，深入思考，认真反思吧。

Richard Hamming --- ``You and Your Research''

At a seminar in the Bell Communications Research Colloquia Series, Dr. Richard W. Hamming, a Professor at the Naval Postgraduate School in Monterey, California and a retired Bell Labs scientist, gave a very interesting and stimulating talk, `You and Your Research' to an overflow audience of some 200 Bellcore staff members and visitors at the Morris Research and Engineering Center on March 7, 1986. This talk centered on Hamming's observations and research on the question ``Why do so few scientists make significant contributions and so many are forgotten in the long run?'' From his more than forty years of experience, thirty of which were at Bell Laboratories, he has made a number of direct observations, asked very pointed questions of scientists about what, how, and why they did things, studied the lives of great scientists and great contributions, and has done introspection and studied theories of creativity. The talk is about what he has learned in

terms of the properties of the individual scientists, their abilities, traits, working habits, attitudes, and philosophy.

> 1986年3月7日，在贝尔通信研究座谈会系列的一次研讨会中，加利福尼亚州蒙特利海军研究生院教授、贝尔实验室退休科学家Richard W. Hamming博士在莫里斯研究与工程中心向大约200名贝尔核心员工和访客进行了一场非常有趣且精彩的演讲`You and Your Research`。本次演讲主要集中于Hamming对于为什么很少有科学家可以做出重大的科学贡献，而大多数科学家都被遗忘在历史长河中这一问题的观察。在他超过40年的经历（其中30年在贝尔实验室）中，他观察了很多，并且对于科学家做什么、如何做以及为什么做提出了非常尖锐的问题，研究伟大科学家的生活和他们的伟大贡献，并且进行了反思且研究了创造力理论。这次演讲是关于他从科学家的个人特性、能力、特质、工作习惯、态度以及哲学方面学习到的内容。

In order to make the information in the talk more widely available, the tape recording that was made of that talk was carefully transcribed. This transcription includes the discussions which followed in the question and answer period. As with any talk, the transcribed version suffers from translation as all the inflections of voice and the gestures of the speaker are lost; one must listen to the tape recording to recapture that part of the presentation. While the recording of Richard Hamming's talk was completely intelligible, that of some of the questioner's remarks were not. Where the tape recording was not intelligible I have added in parentheses my impression of the questioner's remarks. Where there was a question and I could identify the questioner, I have checked with each to ensure the accuracy of my interpretation of their remarks.

> 为了使得演讲中的信息可以更加广泛地获取，这次演讲的录音被认真地进行转录。转录的信息包含问答环节的讨论。与任何谈话一样，转录后的版本将会受到翻译的影响，因为演讲者所有的语调和手势都会丢失，人们必须重新听录像带才能获取演讲的内容。虽然Richard Hamming的演讲是完全可以理解的，但是部分提问者的表达却很难理解。在录音记录中无法理解的部分，我在括号中增加了对于提问者的印象。如果存在问题，我可以确认提问者的身份，并且与他们核实以确保我对于他们言论的解释是准确的。

INTRODUCTION OF DR. RICHARD W. HAMMING

As a speaker in the Bell Communications Research Colloquium Series, Dr. Richard W. Hamming of the Naval Postgraduate School in Monterey, California, was introduced by Alan G. Chynoweth, Vice President, Applied Research, Bell Communications Research.

> 作为贝尔通信研究座谈会系列的发言人，由贝尔通信研究应用研究副总裁Alan G. Chynoweth介绍来自加利福尼亚州蒙特雷海军研究生院的Richard W. Hamming博士。

Alan G. Chynoweth: Greetings colleagues, and also to many of our former colleagues from Bell Labs who, I understand, are here to be with us today on what I regard as a particularly felicitous occasion. It gives me very great pleasure indeed to introduce to you my old friend and colleague from many many years back, Richard Hamming, or Dick Hamming as he has always been know to all of us.

> Alan G. Chynoweth: 向我的同事们以及许多来自贝尔实验室的前同事致意，我认为这是一个特别愉快的时刻。我非常高兴向大家介绍我多年前的老朋友也是我的同事Richard Hamming，或者我们都知道的Dick Hamming。

Dick is one of the all time greats in the mathematics and computer science arenas, as I'm sure the audience here does not need reminding. He received his early education at the Universities of Chicago and Nebraska, and got his Ph.D. at Illinois; he then joined the Los Alamos project during the war. Afterwards, in 1946, he joined Bell Labs. And that is, of course, where I met Dick - when I joined Bell Labs in their physics research organization. In those days, we were in the habit of lunching together as a physics group, and for some reason this strange fellow from mathematics was always pleased to join us. We were always happy to have him with us because he brought so many unorthodox ideas and views. Those lunches were stimulating, I can assure you.

> 正如大家都知道的，Dick是数学和计算机科学领域有史以来最伟大的任务致意。他之前在芝加哥大学和内布拉斯加大学读书，在伊利诺伊州获取博士学位。在战争期间，他加入了洛斯阿拉莫斯项目。之后，在1946年，他加入了贝尔实验室。那就是我和Dick相识的地方，当时我加入了贝尔实验室的物理研究组。在那段时光中，我们经常作为一个物理小组一起吃午饭。不知怎么的，这个喜欢数学的家伙总是喜欢加入我们。我们也非常喜欢和他在一起，因为他能带来一些非常规的想法和观点，我可以向你保证，那些午餐时光非常令人激动。

While our professional paths have not been very close over the years, nevertheless I've always recognized Dick in the halls of Bell Labs and have always had tremendous admiration for what he was doing. I think the record speaks for itself. It is too long to go through all the details, but let me point out, for example, that he has written seven books and of those seven books which tell of various areas of mathematics and computers and coding and information theory, three are already well into their second edition. That is testimony indeed to the prolific output and the stature of Dick Hamming.

> 虽然这些年我们的职业道路不是非常接近，我一直在贝尔实验室的大厅中见到Dick的个人成就，并对他的所作所为充满钦佩。我认为那些记录不言自明。可能了解所有的细节会耗费很长时间，但是我们可以概述一下。他已经写了七本书，讲述了数学、计算机、编码和信息论的许多领域，其中三个已经出版了第二版。这足以证明Dick Hamming的高产成果和地位。

I think I last met him - it must have been about ten years ago - at a rather curious little conference in Dublin, Ireland where we were both speakers. As always, he was tremendously entertaining. Just one more example of the provocative thoughts that he comes up with: I remember him saying, ``There are wavelengths that people cannot see, there are sounds that people cannot hear, and maybe computers have thoughts that people cannot think." Well, with Dick Hamming around, we don't need a computer. I think that we are in for an extremely entertaining talk.

> 我们都曾经在爱尔兰都柏林演讲。和往常一样，他非常有趣。我们可以举一个他提出的激进想法的又一个例子，我记得他说过：“有些波长人类看不到，有些声音人类听不到，也许计算机也会有人类无法思考的想法” 有Dick Hamming，我们甚至不需要电脑了。我认为我们将进行一次非常有趣的谈话。

THE TALK: ``You and Your Research'' by Dr. Richard W. Hamming

It's a pleasure to be here. I doubt if I can live up to the Introduction. The title of my talk is, ``You and Your Research." It is not about managing research, it is about how you individually do your research. I could give a talk on the other subject - but it's not, it's about you. I'm not talking about ordinary run-of-the-mill research; I'm talking about great research. And for the sake of describing great research I'll occasionally say Nobel-Prize type of work. It doesn't have to gain the Nobel Prize, but I mean those kinds of things

which we perceive are significant things. Relativity, if you want, Shannon's information theory, any number of outstanding theories - that's the kind of thing I'm talking about.

> 很高兴来到这里，我担心自己能否达到介绍的要求。我演讲的题目是“你和你的研究”。这不是关于管理研究的，而是关于你个人如何开展你的研究。事实上我可以以另一个话题开展一个演讲，但这次不是，这是关于你个人的。我不是在讨论普普通通的研究，而是在谈论伟大的研究。为了描述伟大的研究，我偶尔会谈到获取诺贝尔奖的研究。但是个人所做的研究不一定要获得诺贝尔奖，只是为了突出这些研究意义重大。相对来说，如果你愿意，Shannon的信息论，任何数目的杰出理论，那大概就是我要讨论的内容。

Now, how did I come to do this study? At Los Alamos I was brought in to run the computing machines which other people had got going, so those scientists and physicists could get back to business. I saw I was a stooge. I saw that although physically I was the same, they were different. And to put the thing bluntly, I was envious. I wanted to know why they were so different from me. I saw Feynman up close. I saw Fermi and Teller. I saw Oppenheimer. I saw Hans Bethe: he was my boss. I saw quite a few very capable people. I became very interested in the difference between those who do and those who might have done.

> Ok，我为什么要开展这项研究呢？在洛斯阿拉莫斯，我被带来运行其他人使用的计算机，这样那些科学家和物理学家就可以重新开始他们的工作了。我发现我就是个配角。虽然在物理层面我们是一样的，但是他们显然有不同之处。坦白说，我很羡慕。我想知道，为什么他们与我如此不同。我近距离见过Feynman，Fermi和Teller。我见到了Oppenheimer，我见到了我的老板Hans Bethe。我见到了很多非常能干的人。我对那些做过这些事和本可以做到这些事的人之间的差别非常感兴趣。

When I came to Bell Labs, I came into a very productive department. Bode was the department head at the time; Shannon was there, and there were other people. I continued examining the questions, `Why?'' and` What is the difference?" I continued subsequently by reading biographies, autobiographies, asking people questions such as: ``How did you come to do this?" I tried to find out what are the differences. And that's what this talk is about.

> 我来到贝尔实验室的时候，我进入了一个非常高效的部门。Bode是当时部门负责人，Shannon也在那里，还有很多其他人。我继续研究这些问题，“为什么”以及“区别是什么？”随后，我开始阅读传记、自传，询问大家一些问题，例如“你是如何做到这一点的？”我试着找出其中的区别。这也是本次演讲的内容。

Now, why is this talk important? I think it is important because, as far as I know, each of you has one life to live. Even if you believe in reincarnation it doesn't do you any good from one life to the next! Why shouldn't you do significant things in this one life, however you define significant? I'm not going to define it - you know what I mean. I will talk mainly about science because that is what I have studied. But so far as I know, and I've been told by others, much of what I say applies to many fields. Outstanding work is characterized very much the same way in most fields, but I will confine myself to science.

> 为什么这些演讲很重要呢？我认为很重要，据我说知，大家都要生活。即使你相信来生，从这一世到下一世不会带来很多益处，为什么不在这一生中做重要的事情呢？无论你是如何定义重要的。我也不会去定义它。我主要讨论科学，因为那是我所研究的。但是据我所知，很多人告诉我，我所说的也可以应用于许多其它领域。在大多数领域中，杰出的工作都有很多相似性，不过我只把自己局限于科学。

In order to get at you individually, I must talk in the first person. I have to get you to drop modesty and say to yourself, `Yes, I would like to do first-class work.'' Our society frowns on people who set out to do really good work. You're not supposed to; luck is supposed to descend on you and you do great things by chance. Well, that's a kind of dumb thing to say. I say, why shouldn't you set out to do something significant. You don't have to tell other people, but shouldn't you say to yourself,` Yes, I would like to do something significant."

> 为了以个人的视角来完成研究，我需要使用第一人称完成演讲。我需要让大家放下谦虚，对自己说“是的，我想做一流的工作” 我们的社会对那些努力做好工作的人感到不满：你不应该那么做，运气会降临在你头上，你只会在偶然的机会中做出伟大的事情。好吧，这是一种愚蠢的说法。如果让我说，为什么你不着手做一些重要的事情呢？你不需要告诉别人，但是你至少应该对自己说：“是的，我想做一些重要的事情。”

In order to get to the second stage, I have to drop modesty and talk in the first person about what I've seen, what I've done, and what I've heard. I'm going to talk about people, some of whom you know, and I trust that when we leave, you won't quote me as saying some of the things I said.

> 为了继续，我需要放下谦虚，以第一人称讨论我所看到的、我所做的以及我所听到的。我会讨论到一些人，其中一些人你们也认识。

Let me start not logically, but psychologically. I find that the major objection is that people think great science is done by luck. It's all a matter of luck. Well, consider Einstein. Note how many different things he did that were good. Was it all luck? Wasn't it a little too repetitive? Consider Shannon. He didn't do just information theory. Several years before, he did some other good things and some which are still locked up in the security of cryptography. He did many good things.

> 我将从心理上而不是逻辑上开始。我发现主要的反对意见是人们认为伟大的科学是靠运气完成，只关于运气。想想爱因斯坦的，他做了很多好工作。这都是运气吗，这概率也太大了。想想Shannon，他不知做了信息论，几年前他还做了其他的事情，不过有些仍然在保密。他的确做了很多伟大的工作。

You see again and again, that it is more than one thing from a good person. Once in a while a person does only one thing in his whole life, and we'll talk about that later, but a lot of times there is repetition. I claim that luck will not cover everything. And I will cite Pasteur who said, ``Luck favors the prepared mind." And I think that says it the way I believe it. There is indeed an element of luck, and no, there isn't. The prepared mind sooner or later finds something important and does it. So yes, it is luck. The particular thing you do is luck, but that you do something is not.

> 你会一次有一次看到，一个优秀的人做了很多事情。偶尔一个人一生只做好一件事，我们稍后会做一些讨论，但是很多时候是具有共性的。我认为运气不会涵盖一切，我引用Pasteur的话：“运气偏爱有准备的头脑”。我认为这就是我所相信的事情。当然一定有运气成分，但是有准备的头脑迟早会发现一些重要的事情并且开始尝试去做。所以，如果你所做的特别的事情是运气，但是你做事情这件事情并不是运气。

For example, when I came to Bell Labs, I shared an office for a while with Shannon. At the same time he was doing information theory, I was doing coding theory. It is suspicious that the two of us did it at the same place and at the same time - it was in the atmosphere. And you can say, `Yes, it was luck.'' On the other hand you can say,` But why of all the people in Bell Labs then were those the two who did it?" Yes, it is partly luck, and partly it is the prepared mind; but `partly' is the other thing I'm going to talk about. So, although I'll come back several more times to luck, I want to dispose of this matter of luck as being the sole criterion whether you do great work or not. I claim you have some, but not total, control over it. And I will quote, finally, Newton on the matter. Newton said, ``If others would think as hard as I did, then they would get similar results."

> 例如，当我来到贝尔实验室时，我与Shannon共用了一间办公室。他在做信息论，而我在做编码理论。在这种氛围喜爱，我们在同一地点、同一时间完成了这两件事。你可以说，“是的，这是运气”。另外，你也可以说：“为什么在贝尔实验室那么多人，只有两个人做了这两件事？”就是这样，一部分是运气，另一部分是时刻准备的内心。但是部分就是我要讨论的另一件事。尽管我会再重新讨论到运气，我想把运气作为你是否做了一个好工作的唯一标准。我认为你对于运气有部分控制权，当然不是全部的控制权。最后，我引用牛顿的话：“如果其他人像我一样认真思考，那么他们也会得到类似的结果”。

One of the characteristics you see, and many people have it including great scientists, is that usually when they were young they had independent thoughts and had the courage to pursue them. For example, Einstein, somewhere around 12 or 14, asked himself the question, ``What would a light wave look like if I went with the velocity of light to look at it?" Now he knew that electromagnetic theory says you cannot have a stationary local maximum. But if he moved along with the velocity of light, he would see a local maximum. He could see a contradiction at the age of 12, 14, or somewhere around there, that everything was not right and that the velocity of light had something peculiar. Is it luck that he finally created special relativity? Early on, he had laid down some of the pieces by thinking of the fragments. Now that's the necessary but not sufficient condition. All of these items I will talk about are both luck and not luck.

> 你可以观察到一个特点，包括伟大科学家在内的很多人都有，就是他们在年轻的时候拥有自己的思想，并且有勇气去追求这些思想。例如，爱因斯坦在12或14岁左右的时候就询问自己的一个问题：“当我以光速观察光波，它将是什么样子？”现在他知道电磁理论说不可能有一个固定的局部最大值。但是，当他以光速移动，他将能够看到一个局部最大值。他在12、14岁左右就可以在某个领域意识到

> 一些矛盾点，即所有事情都不对，光有一些难以理解的事情。他最终创建了狭义相对论，这真的是运气吗？早些时候，他已经通过思考这些问题得到了一些结论。所以运是一个必要但不充分的条件。我将讨论的所有内容都是运气与非运气的混合体。

How about having lots of `brains?' It sounds good. Most of you in this room probably have more than enough brains to do first-class work. But great work is something else than mere brains. Brains are measured in various ways. In mathematics, theoretical physics, astrophysics, typically brains correlates to a great extent with the ability to manipulate symbols. And so the typical IQ test is apt to score them fairly high. On the other hand, in other fields it is something different. For example, Bill Pfann, the fellow who did zone melting, came into my office one day. He had this idea dimly in his mind about what he wanted and he had some equations. It was pretty clear to me that this man didn't know much mathematics and he wasn't really articulate. His problem seemed interesting so I took it home and did a little work. I finally showed him how to run computers so he could compute his own answers. I gave him the power to compute. He went ahead, with negligible recognition from his own department, but ultimately he has collected all the prizes in the field. Once he got well started, his shyness, his awkwardness, his inarticulateness, fell away and he became much more productive in many other ways. Certainly he became much more articulate.

> 那么头脑呢？这听起来不错，在座的各位可能有足够的头脑来做一流的工作。但是伟大的工作不仅仅是头脑。衡量头脑的方法有很多种。在数学、理论物理、天梯物理学，头脑通常与处理符号的能力有关。因此，典型的智商测试会给他们相对高的分数。但是在其它领域则存在不同。例如，有一天，做Zone Melting的Bill Pfann来到我的办公室。他对于自己想得到什么有模糊的想法，并且有一些等式。我直到这个人不太懂数学，也不是非常逻辑清晰。他的问题似乎很有趣，于是我把这些问题带回家并做了一些工作。我最终向他展示如何运行计算机，这样他就可以运行得到自己的答案了。我帮助他进行计算，他继续向前，尽管当时他的部门对他没有足够的认可，但他最终获取了该领域中的所有奖项。一旦他有了良好的开端，他的害羞、笨拙和逻辑不清晰就消失了，他在许多其他方面变得更加有效率。当然，他也变得逻辑更加清晰。

And I can cite another person in the same way. I trust he isn't in the audience, i.e. a fellow named Clogston. I met him when I was working on a problem with John Pierce's group and I didn't think he had much. I asked my friends who had been with him at school, `Was he like that in graduate school?''` Yes," they replied. Well I would have fired

the fellow, but J. R. Pierce was smart and kept him on. Clogston finally did the Clogston cable. After that there was a steady stream of good ideas. One success brought him confidence and courage.

> 我可以以同样的方式引用另一个人。我相信他没在观众席中，就是Clogston。我在与John Pierce的团队处理一个问题时遇到了他，我认为他了解的不多。我问曾经与他一起上学的朋友：“他在研究生院的时候就是这样吗”，“是的”，他们回答到。好吧，我本来要解雇那个假货的，但是J. R. Pierce很聪明， 并留住了他。Clogston最终搞定了Clogston cable。自那之后，他的好的想法源源不断。一次成功带给了他信心和勇气。

One of the characteristics of successful scientists is having courage. Once you get your courage up and believe that you can do important problems, then you can. If you think you can't, almost surely you are not going to. Courage is one of the things that Shannon had supremely. You have only to think of his major theorem. He wants to create a method of coding, but he doesn't know what to do so he makes a random code. Then he is stuck. And then he asks the impossible question, ``What would the average random code do?'' He then proves that the average code is arbitrarily good, and that therefore there must be at least one good code. Who but a man of infinite courage could have dared to think those thoughts? That is the characteristic of great scientists; they have courage. They will go forward under incredible circumstances; they think and continue to think.

> 成功科学家的特点之一就是有勇气。一旦你鼓起勇气并相信你能够解决重要的问题，那么你就可以。如果你认为你做不到，那么几乎可以肯定你做不到。勇气是Shannon拥有的宝物之一。你只需要想想他的主要理论。他想要创建一个编码的方法，但是他不知道应该怎么做，因此他就是创建了一个random code。然后他被卡住了。接下来他问了一个不可能回答的问题：“average random code会做什么？”接着他证明了average code是任意好的，因此必须至少存在一个好的code。只有一个拥有无限勇气的人才能想到这些观点。这就是伟大科学家的特点，他们拥有勇气。他们在令人难以置信的情况下继续前进，他们持之以恒地思考。

Age is another factor which the physicists particularly worry about. They always are saying that you have got to do it when you are young or you will never do it. Einstein did things very early, and all the quantum mechanic fellows were disgustingly young when they did their best work. Most mathematicians, theoretical physicists, and astrophysicists do what we consider their best work when they are young. It is not that they don't do good work in their old age but what we value most is often what they did early. On the other

hand, in music, politics and literature, often what we consider their best work was done late. I don't know how whatever field you are in fits this scale, but age has some effect.

> 年龄是物理学家非常担心的一个因素。他们总是说，你必须在年轻的时候做到这件事情，否则你就永远也做不到。爱因斯坦很早就做到了伟大的工作，而且所有的量子力学研究人员都在年轻的时候就做了很好的工作。大多数数学家、理论物理学家和天体物理学家在年轻时都做了我们认为最好的工作。这并不说他们后续的工作不好，而是我们会更看重他们早期做的工作。另一方面，在音乐、政治和文学方面， 我们认为他们后期做的工作是最好的。我不知道你所在的任何领域是否会符合这个标准，但是年龄可能会有一些影响。

But let me say why age seems to have the effect it does. In the first place if you do some good work you will find yourself on all kinds of committees and unable to do any more work. You may find yourself as I saw Brattain when he got a Nobel Prize. The day the prize was announced we all assembled in Arnold Auditorium; all three winners got up and made speeches. The third one, Brattain, practically with tears in his eyes, said, `I know about this Nobel-Prize effect and I am not going to let it affect me; I am going to remain good old Walter Brattain.'' Well I said to myself,` That is nice." But in a few weeks I saw it was affecting him. Now he could only work on great problems.

> 但是请让我讲述一下为什么年龄似乎会产生这样的影响。首先，一旦你做了一些好的工作，你会发现自己处在各种各样的委员会中，无法再进行任何工作。你可能发现自己就像我看到Brattain获得诺贝尔奖时一样。诺贝尔奖公布的时候我们都在阿诺德礼堂，三位获奖者都发表了演讲。第三个，Brattain，几乎热泪盈眶地说：“我知道诺贝尔奖的影响，但我不会让他影响到我。我永远是老Walte Brattain”。我对自己说：“太好了”。但是几周后，我发现了这对他的影响，现在他只能解决大问题了。

When you are famous it is hard to work on small problems. This is what did Shannon in. After information theory, what do you do for an encore? The great scientists often make this error. They fail to continue to plant the little acorns from which the mighty oak trees grow. They try to get the big thing right off. And that isn't the way things go. So that is another reason why you find that when you get early recognition it seems to sterilize you. In fact I will give you my favorite quotation of many years. The Institute for Advanced Study in Princeton, in my opinion, has ruined more good scientists than any institution has created, judged by what they did before they came and judged by what they did after. Not

that they weren't good afterwards, but they were superb before they got there and were only good afterwards.

> 当你出名的时候，再去尝试解决小问题是非常难的。这就是Shannon所做的事情。在信息论之后，你可以为它做什么？伟大的科学家通常会犯这个错误，他们没能在高大橡树下继续种植小种子。他们试图把大事做好，但是事情并非如此。所以这也是为什么当你得到早期的认可后，它似乎会让你无法继续发展的另一个原因。后面我会给您我这些年非常喜欢的一段话。在我看来，普林斯顿高等研究所毁掉了比其它任何机构更多的好的科学家，这一点我们可以通过他们来之前和来之后的所做的事情对比发现。这并不是说他们后来的工作不够好，但是他们在来这里之前是superb，但是来到这里后只是good。

This brings up the subject, out of order perhaps, of working conditions. What most people think are the best working conditions, are not. Very clearly they are not because people are often most productive when working conditions are bad. One of the better times of the Cambridge Physical Laboratories was when they had practically shacks - they did some of the best physics ever.

> 这提出了工作条件的问题，可能顺序不太多。大多数人认为的最好的工作环境其实并不是最好的。这很明显，因为人们往往在恶劣的工作条件下最具有生产力。剑桥物理实验室最好的时光是真正有棚屋的时候，他们做了一些最好的物理学工作。

I give you a story from my own private life. Early on it became evident to me that Bell Laboratories was not going to give me the conventional acre of programming people to program computing machines in absolute binary. It was clear they weren't going to. But that was the way everybody did it. I could go to the West Coast and get a job with the airplane companies without any trouble, but the exciting people were at Bell Labs and the fellows out there in the airplane companies were not. I thought for a long while about, `Did I want to go or not?'' and I wondered how I could get the best of two possible worlds. I finally said to myself,` Hamming, you think the machines can do practically everything. Why can't you make them write programs?" What appeared at first to me as a defect forced me into automatic programming very early. What appears to be a fault, often, by a change of viewpoint, turns out to be one of the greatest assets you can have. But you are not likely to think that when you first look the thing and say, ``Gee, I'm never going to get enough programmers, so how can I ever do any great programming?"

> 接下来是我个人的故事。早些时候，我知道贝尔实验室不会给我传统的使用绝对二进制编程的方法。显然他们不会这样做。但是实际上每个人都是这样操作的。我可以去西海岸然后在航空公司找一份工作，但是显然那里与贝尔实验室的同事不同。“我想不想去那里？”并且我想知道如何才能在两个可能的世界中取得最好的结果。我最后对自己说“Hamming，你认为机器几乎可以做任何事情。你为什么不能让他们写程序呢？”最初我认为让计算机写程序是一个缺陷，迫使我很早就开始尝试自动变成。通常来说，通过转变观点，看似错误的东西会成为你可能拥有的最大资产之一。但是，当你第一次看到这个东西的时候，你不太可能想到“天啊，我永远都找不到足够的程序员，怎么能做到伟大的编程呢？”

And there are many other stories of the same kind; Grace Hopper has similar ones. I think that if you look carefully you will see that often the great scientists, by turning the problem around a bit, changed a defect to an asset. For example, many scientists when they found they couldn't do a problem finally began to study why not. They then turned it around the other way and said, ``But of course, this is what it is'' and got an important result. So ideal working conditions are very strange. The ones you want aren't always the best ones for you.

> 还有许多其它类似的故事。Grace Hopper也有类似的例子。我认为，如果你仔细观察，你会发现伟大的科学家经常能够扭转问题，把缺陷变成个人资产。例如，当许多科学见发现自己做不到一件事情的时候，他们会最终研究为什么不能。接下来他们会反过来说“当然，这就是事实”并且得到一个重要的结果。所以，理想的工作条件是很难满足的。你想要的并不总是最适合你的。

Now for the matter of drive. You observe that most great scientists have tremendous drive. I worked for ten years with John Tukey at Bell Labs. He had tremendous drive. One day about three or four years after I joined, I discovered that John Tukey was slightly younger than I was. John was a genius and I clearly was not. Well I went storming into Bode's office and said, `How can anybody my age know as much as John Tukey does?'' He leaned back in his chair, put his hands behind his head, grinned slightly, and said,` You would be surprised Hamming, how much you would know if you worked as hard as he did that many years." I simply slunk out of the office!

> 接下来我们谈一下驱动力的事情。你可以观察到，大多数科学家都有巨大的动力。我在贝尔实验室与John Tukey工作了十多年。他有巨大的驱动力。在我加入的三四年后的一天，我发现John Tukey比我年轻一些，他是个天才但我显然不是。我走进Bode的办公室询问“我这个年纪的人怎么能像John Tukey那样知道这么多？”他向后靠在椅子上，双手放在脑后，微微一笑说：“如果你像他那样努力工作那么多年，你会对自己所知道的事情感到惊讶！”我偷偷溜出了办公室。

What Bode was saying was this: ``Knowledge and productivity are like compound interest." Given two people of approximately the same ability and one person who works ten percent more than the other, the latter will more than twice outproduce the former. The more you know, the more you learn; the more you learn, the more you can do; the more you can do, the more the opportunity - it is very much like compound interest. I don't want to give you a rate, but it is a very high rate. Given two people with exactly the same ability, the one person who manages day in and day out to get in one more hour of thinking will be tremendously more productive over a lifetime. I took Bode's remark to heart; I spent a good deal more of my time for some years trying to work a bit harder and I found, in fact, I could get more work done. I don't like to say it in front of my wife, but I did sort of neglect her sometimes; I needed to study. You have to neglect things if you intend to get what you want done. There's no question about this.

> Bode是这样说的：“知识和生产力就像是复利”。假设两个能力大致相同的人，一个人必另一个人多工作10%，后者的产量将是前者的两倍多。你知道的越多，你学到的就越多；你学到的越多，你能做的就越多；你能做的越多，机会就会越多。这很像复利。我不想给你一个概率，但这其实是很高的几率了。如果两个人有完全相同的能力，一个人每天能够多思考一个小时，那么他一生的工作效率就会大大提高。我把Bode的话记在心里。几年以来，我花了很多时间努力工作，事实上，我发现我可以完成更多工作。我不喜欢在我的妻子面前说这些，但有时我的确有点忽视了她，我只是需要学习。如果你想做你希望做的事情，你就一定要忽略一些事情。这是毫无疑问的。

On this matter of drive Edison says, ``Genius is 99% perspiration and 1% inspiration." He may have been exaggerating, but the idea is that solid work, steadily applied, gets you surprisingly far. The steady application of effort with a little bit more work, intelligently applied is what does it. That's the trouble; drive, misapplied, doesn't get you anywhere. I've often wondered why so many of my good friends at Bell Labs who worked as hard or

harder than I did, didn't have so much to show for it. The misapplication of effort is a very serious matter. Just hard work is not enough - it must be applied sensibly.

> 关于驱动力，爱迪生说："天才是99%的汗水和1%的灵感。"他可能有些夸大其词，但他的想法是，扎实的工作，稳步的应用，会让你走得更远。稳定地付出努力，再多做一些工作，明智的应用就是成功的原因。这就是问题所在，无法应用的驱动力不会帮助你做到任何事情。我经常想知道为什么这么多我在贝尔实验室的好朋友都和我一样努力或者更加努力地工作，却没有表现出那么多。不正确的努力是一个非常严重的问题。仅仅努力是不够的，必须合理明智地应用自己的经理。

There's another trait on the side which I want to talk about; that trait is ambiguity. It took me a while to discover its importance. Most people like to believe something is or is not true. Great scientists tolerate ambiguity very well. They believe the theory enough to go ahead; they doubt it enough to notice the errors and faults so they can step forward and create the new replacement theory. If you believe too much you'll never notice the flaws; if you doubt too much you won't get started. It requires a lovely balance. But most great scientists are well aware of why their theories are true and they are also well aware of some slight misfits which don't quite fit and they don't forget it. Darwin writes in his autobiography that he found it necessary to write down every piece of evidence which appeared to contradict his beliefs because otherwise they would disappear from his mind. When you find apparent flaws you've got to be sensitive and keep track of those things, and keep an eye out for how they can be explained or how the theory can be changed to fit them. Those are often the great contributions. Great contributions are rarely done by adding another decimal place. It comes down to an emotional commitment. Most great scientists are completely committed to their problem. Those who don't become committed seldom produce outstanding, first-class work.

> 我想聊一下另一个特征，是模棱两可 ambiguity。我花了一段时间才发现它的重要性。大多数人喜欢相信某些事情是真的或不是真的。伟大的科学家能够很好地容忍歧义。他们相信这一理论足以继续前行。他们对知识有足够的怀疑，并可以注意到错误和失误，从而能够挺身而出，创造新的替代理论。如果你相信得太多，就永远不会注意到缺点；如果你怀疑得很多，你就很难开始。它需要你达到一个平衡。但是大多数伟大的科学家都很清楚为什么他们的理论是正确的，他们也很清楚一些不太合适的小错误，他们不会忘记。达尔文在自传中写道，他发现有必要把每一个似乎与他的信仰相矛盾的证据都写下来，否则这些证据就会从他的脑海中消失。当你发现明显的缺陷时，你必须保持敏感，跟踪这些事情，并留

> 意如何解释它们，或者如何改变理论以适应它们。 这些往往都是伟大的共线。伟大的贡献通常很少是通过再增加一个小数点来完成。这归结为一种情感上的承。大多数伟大的科学家完全致力于他们的问题。那些没有献身精神的人很少能够做出卓越的、一流的工作。

Now again, emotional commitment is not enough. It is a necessary condition apparently. And I think I can tell you the reason why. Everybody who has studied creativity is driven finally to saying, ``creativity comes out of your subconscious." Somehow, suddenly, there it is. It just appears. Well, we know very little about the subconscious; but one thing you are pretty well aware of is that your dreams also come out of your subconscious. And you're aware your dreams are, to a fair extent, a reworking of the experiences of the day. If you are deeply immersed and committed to a topic, day after day after day, your subconscious has nothing to do but work on your problem. And so you wake up one morning, or on some afternoon, and there's the answer. For those who don't get committed to their current problem, the subconscious goofs off on other things and doesn't produce the big result. So the way to manage yourself is that when you have a real important problem you don't let anything else get the center of your attention - you keep your thoughts on the problem. Keep your subconscious starved so it has to work on your problem, so you can sleep peacefully and get the answer in the morning, free.

> 再说一遍，只有情感上的承诺是不够的。这当然是一个必要条件。我想我可以告诉你原因。每个研究过创造力的人最后都会说：“创造力来源于你的潜意识。”不知怎么的，突然间，它出现了。我们对于潜意识只是甚少。但有一件事情你很清楚，那就是你的梦想也来自于你的潜意识。你意识到，在很大程度上，你的梦想就是对你当天经历的重新塑造。如果你日复一日地沉浸在一个话题中，你的潜意识除了解决你的问题之外别无选择。所以在你某天早上或者下午醒来，答案就在眼前了。对于那些没有致力于解决当前问题的人来说，潜意识会在其它无关的事情上，不会产生好的结果。因此，管理自己的方法是，当你有一个真正重要的问题时，不要让其它事情成为你的注意力，把你的想法放在这个问题上，让你的潜意识挨饿，这样它就必须解决你的问题，这样你就可以平静地睡觉，并在早晨得到答案。

Now Alan Chynoweth mentioned that I used to eat at the physics table. I had been eating with the mathematicians and I found out that I already knew a fair amount of mathematics; in fact, I wasn't learning much. The physics table was, as he said, an exciting place, but I think he exaggerated on how much I contributed. It was very interesting to listen to Shockley, Brattain, Bardeen, J. B. Johnson, Ken McKay and other people, and I was

learning a lot. But unfortunately a Nobel Prize came, and a promotion came, and what was left was the dregs. Nobody wanted what was left. Well, there was no use eating with them!

> Alan Chynoweth提到我过去经常去物理组的餐桌上吃饭。我之前经常和数学家们一起吃饭，我发现我已经知道了相当多的数学知识。事实上，我没学到什么。正如他所说，物理组是一个令人兴奋的抵挡，但我认为他夸大了我的贡献。听Shockley、Brattain、Bardeen、J. B. Johnson、Ken McKay和其它人的讨论非常有趣，我学习到了很多。但不幸的是，诺贝尔奖和升职出现了，剩下的就是没啥用的东西。嗯，和他们一起吃饭是没用的。

Over on the other side of the dining hall was a chemistry table. I had worked with one of the fellows, Dave McCall; furthermore he was courting our secretary at the time. I went over and said, `Do you mind if I join you?'' They can't say no, so I started eating with them for a while. And I started asking,` What are the important problems of your field?" And after a week or so, `What important problems are you working on?'' And after some more time I came in one day and said,` If what you are doing is not important, and if you don't think it is going to lead to something important, why are you at Bell Labs working on it?" I wasn't welcomed after that; I had to find somebody else to eat with! That was in the spring.

> 餐厅的另一边是化学组。我曾经与其中的一位同事Dave McCall共事过。此外，当时他正在讨好我们的秘书。我走过去说，“你介意我加入你们吗？”他们不能拒绝，因为我和他们待了一会。我开始问：“你所在领域的重要问题是什么？”大约一周后，“你此时正在处理什么重要的问题？”过了一段时间后我又来了，说：“如果你正在做的事情不重要，并且你也不认为它能得到一些重要的结果，那么你为什么要在贝尔实验室工作？”在那之后，我就不受欢迎了。我不得不找其他人一起吃饭！那是在春天。

In the fall, Dave McCall stopped me in the hall and said, `Hamming, that remark of yours got underneath my skin. I thought about it all summer, i.e. what were the important problems in my field. I haven't changed my research,'' he says,` but I think it was well worthwhile." And I said, `Thank you Dave,'' and went on. I noticed a couple of months later he was made the head of the department. I noticed the other day he was a Member of the National Academy of Engineering. I noticed he has succeeded. I have never heard the names of any of the other fellows at that table mentioned in science and scientific circles.

`They were unable to ask themselves,` What are the important problems in my field?"

> 秋天，Dave McCall在大厅中拦住我说“Hamming，你的那句话那我大吃一惊。我整个夏天都在思考这个问题，即我所在领域的重要问题是什么？我没有改变我的研究，但我认为那是值得的。”我说“谢谢你Dave”。我注意到，几个月后，他被任命为部门负责人。前几天他成为了美国国家工程院原始。我注意到他成功了。在曾经的那个餐桌上，我从未在科学界见到其他人的名字。他们没有询问自己，“我所在领域的重要问题是什么？”

If you do not work on an important problem, it's unlikely you'll do important work. It's perfectly obvious. Great scientists have thought through, in a careful way, a number of important problems in their field, and they keep an eye on wondering how to attack them. Let me warn you, `important problem' must be phrased carefully. The three outstanding problems in physics, in a certain sense, were never worked on while I was at Bell Labs. By important I mean guaranteed a Nobel Prize and any sum of money you want to mention. We didn't work on (1) time travel, (2) teleportation, and (3) antigravity. They are not important problems because we do not have an attack. It's not the consequence that makes a problem important, it is that you have a reasonable attack. That is what makes a problem important. When I say that most scientists don't work on important problems, I mean it in that sense. The average scientist, so far as I can make out, spends almost all his time working on problems which he believes will not be important and he also doesn't believe that they will lead to important problems.

> 如果你不处理一个重要的问题，你就不太可能做重要的工作。这是非常明显的。伟大的科学家总是以谨慎的方式思考了他们所在领域中的许多重要问题，他们一直关注如何解决这些问题。让我来警告你们，重要问题的措辞必须谨慎。从某种意义上说，物理学的三个突出问题在我在贝尔实验室的时候从没有被解决过。所谓的重要是保证获得诺贝尔奖以及金钱。我们没有研究（1）时间旅行，（2）传输，（3）反重力。它们不是重要的问题因为我们没有办法攻击。让问题变得重要不是结果，而是需要一个合理的攻击方法。这就是使得问题变得重要的方法。这也是我所说的大多数科学家研究的都不是重要的问题。据我所知，普通科学家把问题都花在了他认为不重要的问题上，他也不相信这些会导致严重的问题。

I spoke earlier about planting acorns so that oaks will grow. You can't always know exactly where to be, but you can keep active in places where something might happen. And even if you believe that great science is a matter of luck, you can stand on a mountain top where lightning strikes; you don't have to hide in the valley where you're safe. But the average scientist does routine safe work almost all the time and so he (or she) doesn't produce much. It's that simple. If you want to do great work, you clearly must work on important problems, and you should have an idea.

> 我早些时候谈到种植橡子可以使得橡树生长。你不可能总是知道应该去哪里，但是你可以再可能发生事情的地方保持活跃。即使你相信伟大的科学是运气，你也可以站在可能雷击的山顶上，你不必躲在安全的山谷当中。但是普通的科学家一直在做例行的安全工作，所以他不会产生太多成果。就这么简单。如果你想做伟大的工作，你显然必须解决重要的问题，而且你应该有一个想法。

Along those lines at some urging from John Tukey and others, I finally adopted what I called `Great Thoughts Time.'' When I went to lunch Friday noon, I would only discuss great thoughts after that. By great thoughts I mean ones like:` What will be the role of computers in all of AT&T?", `How will computers change science?'' For example, I came up with the observation at that time that nine out of ten experiments were done in the lab and one in ten on the computer. I made a remark to the vice presidents one time, that it would be reversed, i.e. nine out of ten experiments would be done on the computer and one in ten in the lab. They knew I was a crazy mathematician and had no sense of reality. I knew they were wrong and they've been proved wrong while I have been proved right. They built laboratories when they didn't need them. I saw that computers were transforming science because I spent a lot of time asking` What will be the impact of computers on science and how can I change it?" I asked myself, ``How is it going to change Bell Labs?" I remarked one time, in the same address, that more than one-half of the people at Bell Labs will be interacting closely with computing machines before I leave. Well, you all have terminals now. I thought hard about where was my field going, where were the opportunities, and what were the important things to do. Let me go there so there is a chance I can do important things.

> 在John Turkey和其他人的督促下，我终于采用了我所说的“伟大思想时代”。当我周五中午去吃午饭时，我只会在那之后讨论一些伟大的想法。我所说的伟大思想例如：“计算机在整个AT&T中将扮演什么角色？”“计算机将如何改变科学？”例如，我当时得出的观察结果是，十分之九的实验是在实验室中进行的，十分之一是在计算机中进行的。我有一次对副总裁说，这将会发生逆转，十分之九的实验在计算机中进行，十分之一的实验在实验室中进行。他们知道我是一个疯狂的数学家，对现实毫无概念。我知道他们是错的，他们被证明是错误的，而我被证明是正确的。他们在不需要的时候建造了实验室。我看到计算机正在改变科学，因为我花了很多时间问：“计算机会对科学产生什么影响，我该如何改变它？”我问自己，“这将如何改变贝尔实验室？”有一次，我在同一个地点说，在我离开之前，贝尔实验室将有一半以上的人与计算机密切互动。你们现在都有终端了。我认证思考了我的领域将会奔向何处，机会在哪里，重要的事情是什么。让我去那里，这样我就有机会做重要的事情。

Most great scientists know many important problems. They have something between 10 and 20 important problems for which they are looking for an attack. And when they see a new idea come up, one hears them say `Well that bears on this problem.'' They drop all the other things and get after it. Now I can tell you a horror story that was told to me but I can't vouch for the truth of it. I was sitting in an airport talking to a friend of mine from Los Alamos about how it was lucky that the fission experiment occurred over in Europe when it did because that got us working on the atomic bomb here in the US. He said` No; at Berkeley we had gathered a bunch of data; we didn't get around to reducing it because we were building some more equipment, but if we had reduced that data we would have found fission." They had it in their hands and they didn't pursue it. They came in second!

> 许多伟大科学家都知道很多重要的问题。他们有10到20个重要问题，他们正在寻求攻击。当他们看到一个新的想法出现时，人们会听到他们说：“好吧，这与这个问题有关。”他们会放下所有的其它事情然后追上这个问题。我现在可以告诉你们一个关于我的恐怖故事，但我无法保证它的真实性。我当时坐在机场里，和我一个来自洛斯阿拉莫斯的朋友谈论裂变实验发生在欧洲是多么幸运。因为这让我们在美国研究原子弹。他说：“不，在伯克利我们已经收集到了一堆数据。我们没有抽出时间来处理它，因为我们在建造更多设备，但是如果我们成功处理了这些数据，我们就会发现裂变并掌握他们。”显然，他们没有追求到它，他们只能位居第二。

The great scientists, when an opportunity opens up, get after it and they pursue it. They drop all other things. They get rid of other things and they get after an idea because they had already thought the thing through. Their minds are prepared; they see the opportunity and they go after it. Now of course lots of times it doesn't work out, but you don't have to hit many of them to do some great science. It's kind of easy. One of the chief tricks is to live a long time!

> 伟大的科学家总是在一个机会出现时，抓住它并追求它。他们放弃了所有其它的东西，追求一个想法，因为他们已经把这些事情想清楚了。他们的思想是有准备的。他们看到了机会，就去追求。当然，很多时候都没有成功。但是你不必为了做一些伟大的科学就打击他们。主要的诀窍之一就是长寿。

Another trait, it took me a while to notice. I noticed the following facts about people who work with the door open or the door closed. I notice that if you have the door to your office closed, you get more work done today and tomorrow, and you are more productive than most. But 10 years later somehow you don't know quite know what problems are worth working on; all the hard work you do is sort of tangential in importance. He who works with the door open gets all kinds of interruptions, but he also occasionally gets clues as to what the world is and what might be important. Now I cannot prove the cause and effect sequence because you might say, ``The closed door is symbolic of a closed mind.'' I don't know. But I can say there is a pretty good correlation between those who work with the doors open and those who ultimately do important things, although people who work with doors closed often work harder. Somehow they seem to work on slightly the wrong thing - not much, but enough that they miss fame.

> 另一个特点，我们花了一段时间才注意到，就是关于那些开门和关门工作的人的事实。我注意到，如果你关上办公室的门，今天和明天你将可以完成更多的工作，而且你比大多数人都有效率。但是十年后，不知何故，你不知道什么问题值得解决。你所做的所有艰苦工作都有点无关紧要。敞开大门工作的人会受到各种各样的干扰，但他偶尔也会得到关于世界是什么以及什么可能是重要的一些线索。现在我无法证明因果顺序，因为你可能会说“紧闭的门代表着封闭的思想。”我不知道，但我可以说，那些敞开大门工作的人和那些最终做到重要事情的人之间有很好的相关性，尽管那些关着门工作的人往往工作地更加努力。不知怎么的，他们似乎在做一件有些错误的事情。不是很多，但足以让他们错过名声。

I want to talk on another topic. It is based on the song which I think many of you know, `It ain't what you do, it's the way that you do it.'' I'll start with an example of my own. I was conned into doing on a digital computer, in the absolute binary days, a problem which the best analog computers couldn't do. And I was getting an answer. When I thought carefully and said to myself,` You know, Hamming, you're going to have to file a report on this military job; after you spend a lot of money you're going to have to account for it and every analog installation is going to want the report to see if they can't find flaws in it." I was doing the required integration by a rather crummy method, to say the least, but I was getting the answer. And I realized that in truth the problem was not just to get the answer; it was to demonstrate for the first time, and beyond question, that I could beat the analog computer on its own ground with a digital machine. I reworked the method of solution, created a theory which was nice and elegant, and changed the way we computed the answer; the results were no different. The published report had an elegant method which was later known for years as ``Hamming's Method of Integrating Differential Equations." It is somewhat obsolete now, but for a while it was a very good method. By changing the problem slightly, I did important work rather than trivial work.

> 我想谈论另一个话题，这是一个大家耳熟能详的歌曲：“这不是你要做什么，而是你做事情的方式。”我被要求在数字计算机中做一个最好的模拟计算机无法做的问题。我得到了答案。当我仔细思考并对自己说：“你知道，Hamming，你必须对这个军事工作提交一份报告；在你花了很多钱之后，你将不得不考虑它；每个模拟安装都会希望报告看他们能否发现其中的缺陷。”至少可以说，我用一种相当糟糕的方法进行了所需的集成，但我最终得到了答案。这是我第一次证明，毫无疑问，我可以使用数字机器在自己的基础上击败模拟计算机。我重新设计了求解方案，创建了一个漂亮而优雅的理论，并改变了我们计算答案的方式，结果没有什么不同。但是发表的报告有一种优雅的方法，后面被称为“汉明积分微分方程的方法”，它现在有些过时了，但有一段时间它是一个非常好的方法。通过稍微改变问题，我做了重要的工作，而不是琐碎的工作。

In the same way, when using the machine up in the attic in the early days, I was solving one problem after another after another; a fair number were successful and there were a few failures. I went home one Friday after finishing a problem, and curiously enough I wasn't happy; I was depressed. I could see life being a long sequence of one problem after another after another. After quite a while of thinking I decided, `No, I should be in`

`the mass production of a variable product. I should be concerned with *all* of next year's problems, not just the one in front of my face.'' By changing the question I still got the same kind of results or better, but I changed things and did important work. I attacked the major problem - How do I conquer machines and do all of next year's problems when I don't know what they are going to be? How do I prepare for it? How do I do this one so I'll be on top of it? How do I obey Newton's rule? He said,` If I have seen further than others, it is because I've stood on the shoulders of giants." These days we stand on each other's feet!

> 同样，早期在阁楼上使用机器时，我正在解决一个又一个问题。相当多的问题被成功解决，也有几个失败了。一个星期五，我解决完一个问题回家了，奇怪的是，我并不高兴，我很沮丧。我可以看到生活就是一个又一个问题的序列。经过很长时间的思考后，我决定“不，我应该从事一种可变产品的大规模生产。我应该关心明年所有的问题，而不仅仅是眼前的问题。”通过改变问题，我仍然得到了相同或更好的结果，但我改变了事情，做了重要的工作。我攻击了主要的问题，当我不知道问题是什么的时候我将如何征服机器并解决它们？我该如何准备？我该如何做到这一点，这样我才能掌握它？我该如何遵守牛顿法则？他说：“如果我比其他人看得更远，那是因为我站在巨人的肩膀上。”这些天我们互相支持。

You should do your job in such a fashion that others can build on top of it, so they will indeed say, ``Yes, I've stood on so and so's shoulders and I saw further." The essence of science is cumulative. By changing a problem slightly you can often do great work rather than merely good work. Instead of attacking isolated problems, I made the resolution that I would never again solve an isolated problem except as characteristic of a class.

> 你应该以这样一种方式来完成你的工作，让其他人可以在此基础上再接再厉，所以他们确实会说，“是的，我站在某人的肩膀上，我看得更远。”科学的本质是累积的，通过稍微改变一个问题，你通常可以做得很好，而不仅仅是做得好。我没有攻击鼓励的问题，我下定决心，除非可以作为一类特征，否则我永远不会再解决鼓励的问题。

Now if you are much of a mathematician you know that the effort to generalize often means that the solution is simple. Often by stopping and saying, ``This is the problem he wants but this is characteristic of so and so. Yes, I can attack the whole class with a far superior method than the particular one because I was earlier embedded in needless detail." The business of abstraction frequently makes things simple. Furthermore, I filed away the methods and prepared for the future problems.

> 现在，如果你是一个数学家，你知道归纳的精力通常意味着解决方案很简单。经常停下来说："这是他想要的问题，但这是其它部分的特点。是的，我可以用比这个特定方法更优越的方法攻击这一类，因为我之前嵌入了不必要的细节。"归纳的工作通常使得工作更加简单。此外，我归纳了这些方法，并且为未来的问题做出了准备。

To end this part, I'll remind you, ``It is a poor workman who blames his tools - the good man gets on with the job, given what he's got, and gets the best answer he can." And I suggest that by altering the problem, by looking at the thing differently, you can make a great deal of difference in your final productivity because you can either do it in such a fashion that people can indeed build on what you've done, or you can do it in such a fashion that the next person has to essentially duplicate again what you've done. It isn't just a matter of the job, it's the way you write the report, the way you write the paper, the whole attitude. It's just as easy to do a broad, general job as one very special case. And it's much more satisfying and rewarding!

> 在这个部分的结尾，我要提醒你："一个可怜的工人才会埋怨自己的工具，一个优秀的人会继续工作，在他拥有的基础上，得到最好的答案。"我建议，通过改变问题，以不同的方式看待问题，你可以对你的最终生产力产生很大的影响，因为你可以以这样一种方式来做，人们确实可以在你所做的基础上再接再厉，或者你可以以下一个人必须再次复制你所做的方式来做。这不仅仅是工作的问题，而是你写报告的方式，你写论文的方式，以及整个过程的态度。做一个广泛且通用的工作与在特殊场景中做一个工作是同样容易的，但是它更令人满意，更有收获。

I have now come down to a topic which is very distasteful; it is not sufficient to do a job, you have to sell it. `Selling' to a scientist is an awkward thing to do. It's very ugly; you shouldn't have to do it. The world is supposed to be waiting, and when you do something great, they should rush out and welcome it. But the fact is everyone is busy with their own work. You must present it so well that they will set aside what they are doing, look at what

you've done, read it, and come back and say, ``Yes, that was good." I suggest that when you open a journal, as you turn the pages, you ask why you read some articles and not others. You had better write your report so when it is published in the Physical Review, or wherever else you want it, as the readers are turning the pages they won't just turn your pages but they will stop and read yours. If they don't stop and read it, you won't get credit.

> 我现在谈一个非常令人反感的话题，仅仅做一项工作是不够的，你需要把它卖出去。对于科学家来说，把东西卖出去是一个尴尬的事情。折痕抽罗，你不应该这样做。整个世界都在等待，当你做了一件伟大的事情时，他们应该欢迎他。但是事实是，每个人都在忙于自己的工作，你必须把它表现地很好，让他们把正在做的事情放在一边，看看你做了什么，读一下，然后说："是的，那很好。"我建议，当你打开一本期刊时，在你翻页的时候，你会问为什么你在读某些文章而不是其他文章。你最好写下你的报告，这样当它发表在《Physical Review》或其它期刊时，当读者翻页时，他们不仅会翻页，还会停下来阅读你的报告。如果他们不停下来读一读，你就不会受到表扬。

There are three things you have to do in selling. You have to learn to write clearly and well so that people will read it, you must learn to give reasonably formal talks, and you also must learn to give informal talks. We had a lot of so-called `back room scientists.' In a conference, they would keep quiet. Three weeks later after a decision was made they filed a report saying why you should do so and so. Well, it was too late. They would not stand up right in the middle of a hot conference, in the middle of activity, and say, ``We should do this for these reasons." You need to master that form of communication as well as prepared speeches.

> 在卖出去的过程中，你必须能够做到三件事。你必须学会写得清晰而且好，这样人们才能读到他；你必须学会合理的正式演讲，也必须学会非正式的演讲。我们有很多所谓的"幕后科学家"在会议中保持沉默。三周后，在做出决定后，他们提交了一份报告，说明你为什么应该这样做。但是为时已晚。他们不会再一场激烈的会议、活动中站起来说："出于这些原因，我们应该这样做。"你需要掌握这种沟通方式以及事先准备好演讲。

When I first started, I got practically physically ill while giving a speech, and I was very, very nervous. I realized I either had to learn to give speeches smoothly or I would essentially partially cripple my whole career. The first time IBM asked me to give a speech in New York one evening, I decided I was going to give a really good speech, a speech that was wanted, not a technical one but a broad one, and at the end if they liked it, I'd

quietly say, ``Any time you want one I'll come in and give you one." As a result, I got a great deal of practice giving speeches to a limited audience and I got over being afraid. Furthermore, I could also then study what methods were effective and what were ineffective.

> 当我开始演讲时，我会产生身体不适，非常紧张。我意识到，我要买必须学会顺利地发表演讲，要么我的整个职业生涯都将部分瘫痪。一天晚上，当IBM第一次邀请我在纽约发表演讲时，我决定要发表一场非常好的演讲，一场人们希望听到的演讲，不是技术性的，而是一个通用的。最后，如果他们喜欢，我会悄悄说，任何时候你想要，我都会来做一个这样的演讲。因此，我在有限的听众面前进行了大量的演讲练习，我克服了恐惧。此外，我还可以研究哪些方法有效，哪些方法无效。

While going to meetings I had already been studying why some papers are remembered and most are not. The technical person wants to give a highly limited technical talk. Most of the time the audience wants a broad general talk and wants much more survey and background than the speaker is willing to give. As a result, many talks are ineffective. The speaker names a topic and suddenly plunges into the details he's solved. Few people in the audience may follow. You should paint a general picture to say why it's important, and then slowly give a sketch of what was done. Then a larger number of people will say, `Yes, Joe has done that,'' or` Mary has done that; I really see where it is; yes, Mary really gave a good talk; I understand what Mary has done." The tendency is to give a highly restricted, safe talk; this is usually ineffective. Furthermore, many talks are filled with far too much information. So I say this idea of selling is obvious.

> 在参加会议时，我已经在研究为什么有些论文被记住，而大多数却没有。技术人员希望进行高度有限的技术性演讲。大多数时候，听众想要听一个通常的演讲，想要比演讲者所讲述的更多的研究和背景。因此，许多演讲都是无效的。演讲者说出一个话题，却突然陷入他要解决的细节中。听众中可能很少有人跟随。你应该画一副全局概览的图，说明为什么它很重要，然后慢慢地画出所做的事情的草图。然后，更多的人说：“是的，Joe那样做了”或“Mary那样做了；我真的明白了它在说什么；是的，Mary真得讲得很好；我理解Mary做的事情。”如果倾向于发表一个高度受限、安全的演讲，这通常是无效的。此外，许多演讲都包含了太多的信息，这种把想法卖出去的想法是显而易见的。

Let me summarize. You've got to work on important problems. I deny that it is all luck, but I admit there is a fair element of luck. I subscribe to Pasteur's `Luck favors the prepared mind.'' I favor heavily what I did. Friday afternoons for years - great thoughts only - means that I committed 10% of my time trying to understand the bigger problems in the field, i.e. what was and what was not important. I found in the early days I had believed this' and yet had spent all week marching in `that' direction. It was kind of foolish. If I really believe the action is over there, why do I march in this direction? I either had to change my goal or change what I did. So I changed something I did and I marched in the direction I thought was important. It's that easy.

> 让我总结一下。你必须解决重要的问题。我否认这都是运气，但是我承认运气也是一个成分。我赞同Pasteur的“运气偏爱有准备的头脑。”我非常赞成我自己所做的事情。多年来的周五下午，我都花出10%的想法试图理解这个领域中更大的问题，即什么问题是重要的，什么不是重要的。早期，我相信了一件事情，但是却花了整整一周时间朝着另一个方向前进。这有点愚蠢。如果我真的相信应该走向一方，为什么要朝着另一个方向前进？我要么改变我的目标，要么改变我所做的。所以我改变了我所做的事情，朝着我认为重要的方向前进，就这么简单。

Now you might tell me you haven't got control over what you have to work on. Well, when you first begin, you may not. But once you're moderately successful, there are more people asking for results than you can deliver and you have some power of choice, but not completely. I'll tell you a story about that, and it bears on the subject of educating your boss. I had a boss named Schelkunoff; he was, and still is, a very good friend of mine. Some military person came to me and demanded some answers by Friday. Well, I had already dedicated my computing resources to reducing data on the fly for a group of scientists; I was knee deep in short, small, important problems. This military person wanted me to solve his problem by the end of the day on Friday. I said, `No, I'll give it to you Monday. I can work on it over the weekend. I'm not going to do it now.'' He goes down to my boss, Schelkunoff, and Schelkunoff says, You must run this for him; he's got to have it by Friday." I tell him, `Why do I?''; he says, You have to." I said, `Fine, Sergei, but you're sitting in your office Friday afternoon catching the late bus home to watch as this fellow walks out that door.'' I gave the military person the answers late Friday afternoon. I then went to Schelkunoff's office and sat down; as the man goes out I say, You

see Schelkunoff, this fellow has nothing under his arm; but I gave him the answers." On Monday morning Schelkunoff called him up and said, `Did you come in to work over the weekend?'' I could hear, as it were, a pause as the fellow ran through his mind of what was going to happen; but he knew he would have had to sign in, and he'd better not say he had when he hadn't, so he said he hadn't. Ever after that Schelkunoff said,` You set your deadlines; you can change them."

> 你可能会告诉我，你还不能控制自己必须做什么。当你刚开始的时候，你可能不会。但一旦你取得了适度的成果，很多人要求的结果就比你能提供的更多，你也有一些选择的权力，但不是完全掌控。我给你讲一个关于这方面的故事，它涉及到教育你的老板。我的一个老板叫Schelkunoff，他一直是我的好朋友。一些军人找到我要求在周五之前得到答案。我已经把我的计算资源用于为一群科学家去除无用的数据，而且我陷入了一些琐碎且重要的问题。这位军人希望我能在周五结束前解决他的问题。我说："不，我会在下周一给你。我可以在周末把它处理好，但我现在不会做。"他去找我的老板，Schelkunoff说："你必须为他做这件事情，他需要在周五之前拿到结果。""我为什么要这样做？"他说："你必须得这样做。""好吧。但是你需要周五下午在办公室等着，直到那位军人走出去。"周五下午晚些时候，我给了这位军人结果，然后我去了Schelkunoff的办公室，坐了下来，当那位军人走出去之后，我说："你看，Schelkunoff，这家伙啥也没有带走，但是我给了他结果。"周一早晨，Schelkunoff给他打电话说："你周末来工作了吗？"我能听到，那家伙停顿了一下，最后说没有。自那之后，Schelkunoff说："你设置了DDL，但是你可以改变它们。"

One lesson was sufficient to educate my boss as to why I didn't want to do big jobs that displaced exploratory research and why I was justified in not doing crash jobs which absorb all the research computing facilities. I wanted instead to use the facilities to compute a large number of small problems. Again, in the early days, I was limited in computing capacity and it was clear, in my area, that a `mathematician had no use for machines.'' But I needed more machine capacity. Every time I had to tell some scientist in some other area,` No I can't; I haven't the machine capacity," he complained. I said `Go tell *your* Vice President that Hamming needs more computing capacity.'' After a while I could see what was happening up there at the top; many people said to my Vice President,` Your man needs more computing capacity." I got it!

> 一次教训足以告诉我的老板，为什么我不想做取代探索性研究的大工作，为什么我有理由不做吸收所有研究计算设施的临时工作。相反，我想使用这些设施来计算大量的小问题。同样，在早期，我的计算能力有限，而且很明显在我的领域“数学家不需要机器。”但是我需要更多的机器容量。每次我都需要告诉其它领域的科学家：“我做不到，我没有足够的机器容量”，他们都会抱怨。我说：“去告诉你的副总裁，Hamming需要更强大的计算能力。”过了一阵子，我可以看到发生了什么，许多人对我的副总裁说：“你的员工需要更多计算能力。”然后我就拥有了更强的计算能力。

I also did a second thing. When I loaned what little programming power we had to help in the early days of computing, I said, `We are not getting the recognition for our programmers that they deserve. When you publish a paper you will thank that programmer or you aren't getting any more help from me. That programmer is going to be thanked by name; she's worked hard.'' I waited a couple of years. I then went through a year of BSTJ articles and counted what fraction thanked some programmer. I took it into the boss and said,` That's the central role computing is playing in Bell Labs; if the BSTJ is important, that's how important computing is." He had to give in. You can educate your bosses. It's a hard job. In this talk I'm only viewing from the bottom up; I'm not viewing from the top down. But I am telling you how you can get what you want in spite of top management. You have to sell your ideas there also.

> 做了另外一件事。当我借用我们在计算早期的一点点编程能力时，我说：“我们的程序员没有得到应有的认可。当你发表一篇论文，你会感谢那个程序员或者你没有得到我更多的帮助。她应该被致谢，她工作非常努力。”我等待了很多年，然后我看了一年的BSTJ的论文并且数了数感谢程序员的份数。我把它那给老板说：“这就是计算在贝尔实验室中所扮演的核心角色；如果BSTJ是重要的，那么计算就是这么重要。”老板不得不让步，你可以借机教育你的老板。在这次演讲中，我只是从下往上看，而不是从上往下搬。但是我告诉你，尽管有高层管理你应该如何才能得到你想要的。你也必须在那里推销你的想法。

Well I now come down to the topic, `Is the effort to be a great scientist worth it?'' To answer this, you must ask people. When you get beyond their modesty, most people will say,` Yes, doing really first-class work, and knowing it, is as good as wine, women and song put together," or if it's a

woman she says, ``It is as good as wine, men and song put together." And if you look at the bosses, they tend to come back or ask for reports, trying to participate in those moments of discovery. They're always in the way. So evidently those who have done it, want to do it again. But it is a limited survey. I have never dared to go out and ask those who didn't do great work how they felt about the matter. It's a biased sample, but I still think it is worth the struggle. I think it is very definitely worth the struggle to try and do first-class work because the truth is, the value is in the struggle more than it is in the result. The struggle to make something of yourself seems to be worthwhile in itself. The success and fame are sort of dividends, in my opinion.

> OK，我们接下来聊下一个话题：“为了成为一位伟大科学家需要付出的努力值得吗？”要回答这个问题，你必须问其他人。当你让他们不再刻意谦虚，大多数人会说：“是的，做真正一流的工作，并且了解它，就像把红酒、爱人和歌曲放在一起一样美妙。”如果你看看那些老板，他们往往会返回来并要求你做报告，试图参与那些新发现的时刻。他们总是会挡路。但是事实证明那些做过这种事的人会还想再做一次。但这是一项有限的调查，我从来不敢出去问那些工作不出色的人对于这件事的感受，这是一个有偏见的样本，但我仍然认为值得努力。我认为努力做一流的工作绝对值得，因为事实是，价值在于斗争的过程，而不是那个结果。努力让自己有所成就本身似乎就是值得的，在我看来，成功和名声是一种红利。

I've told you how to do it. It is so easy, so why do so many people, with all their talents, fail? For example, my opinion, to this day, is that there are in the mathematics department at Bell Labs quite a few people far more able and far better endowed than I, but they didn't produce as much. Some of them did produce more than I did; Shannon produced more than I did, and some others produced a lot, but I was highly productive against a lot of other fellows who were better equipped. Why is it so? What happened to them? Why do so many of the people who have great promise, fail?

> 我已经告诉你如何做到了。这非常简单，为什么这么多才华横溢的人会失败呢？例如，直到今天，我的观点是，在贝尔实验室的数学学院中，有相当多的人比我更有能力、更有天赋，但是他们没有产生那么多成果。他们中的一些人确实比我产出更多，Shannon比我产出更多，其他人的产出也很多，但是我在与其他更优秀的人的比赛中表现得很好。为什么会这样？他们怎么了？为什么有这么多有前途的人都失败了。

Well, one of the reasons is drive and commitment. The people who do great work with less ability but who are committed to it, get more done that those who have great skill and dabble in it, who work during the day and go home and do other things and come back and work the next day. They don't have the deep commitment that is apparently necessary for really first-class work. They turn out lots of good work, but we were talking, remember, about first-class work. There is a difference. Good people, very talented people, almost always turn out good work. We're talking about the outstanding work, the type of work that gets the Nobel Prize and gets recognition.

> 原因之一是动力和承诺。那些能力较弱但是致力于工作的人，比那些拥有高超技术但是只是偶尔做事情的人做得更多，他们白天工作，回家后做其它事情，第二天回来再工作。他们没有做真正一流工作所需要的坚定决心。他们做出了很多好的工作，但是请记住我们在讨论一流的工作，这是有区别的。一个优秀的人，非常有才华的人，几乎总是能够好的工作。我们在讨论杰出的工作，获取诺贝尔奖并得到认可的作品类型。

The second thing is, I think, the problem of personality defects. Now I'll cite a fellow whom I met out in Irvine. He had been the head of a computing center and he was temporarily on assignment as a special assistant to the president of the university. It was obvious he had a job with a great future. He took me into his office one time and showed me his method of getting letters done and how he took care of his correspondence. He pointed out how inefficient the secretary was. He kept all his letters stacked around there; he knew where everything was. And he would, on his word processor, get the letter out. He was bragging how marvelous it was and how he could get so much more work done without the secretary's interference. Well, behind his back, I talked to the secretary. The secretary said, `Of course I can't help him; I don't get his mail. He won't give me the stuff to log in; I don't know where he puts it on the floor. Of course I can't help him.'' So I went to him and said,` Look, if you adopt the present method and do what you can do single-handedly, you can go just that far and no farther than you can do single-handedly. If you will learn to work with the system, you can go as far as the system will support you." And, he never went any further. He had his personality defect of wanting total control and was not willing to recognize that you need the support of the system.

> 第二件事情是人格缺陷的问题。我需要举一个我在Irvine遇到的人。他曾经是一家计算中心的负责人，被临时指派为该大学校长的特别助理。很明显，他有一份前途光明的工作。有一次，他带我去了他的办公室，向我展示了他写信的方法以及如何处理新建。他指出秘书的工作效率有多低。他把所有的信件都堆积在那里；他知道所有的一切都在那里。他会用文字处理器把信件拿出来。他在吹嘘这是多么了不起，他是如何在没有秘书的帮助的情况下完成这么多工作。嗯，之后我和秘书谈过。秘书说："我当然帮助不了他，我没有收到他的邮件。他不会让我去处理，我也不知道他把那些信件放在哪里了。我当然帮助不了他"。我和这个特别助理说，"你看，如果你采用现在的方法，独自做你能做的事情，你就可以走得和独自走一样远，但是如果你学会了使用这个系统，你就可以让系统帮助你"。他没有走得更远。因为他有想要完全控制的人格缺陷，不愿意承认你需要系统的支持。"

You find this happening again and again; good scientists will fight the system rather than learn to work with the system and take advantage of all the system has to offer. It has a lot, if you learn how to use it. It takes patience, but you can learn how to use the system pretty well, and you can learn how to get around it. After all, if you want a decision `No', you just go to your boss and get a` No' easy. If you want to do something, don't ask, do it. Present him with an accomplished fact. Don't give him a chance to tell you `No'. But if you want a` No', it's easy to get a `No'.

> 你会发现这种情况一次又一次地发生，优秀的科学家会与系统抗争而不是学会使用系统工作，并利用系统提供的一切。如果你学会了如何使用它，它能做到很多。这需要耐心，你可以学会很好地使用这个系统，也可以学会如何绕过它。毕竟，如果你想做出否定的决定，你可以找到你的老板然后得到一个否定的答案。如果你想做某件事情，不要问，就去做。给他一个已经完成的事实。不要给他机会对你说"不"。如果你想得到一个否定的答案，你会很容易就得到一个否定的答案。

Another personality defect is ego assertion and I'll speak in this case of my own experience. I came from Los Alamos and in the early days I was using a machine in New York at 590 Madison Avenue where we merely rented time. I was still dressing in western clothes, big slash pockets, a bolo and all those things. I vaguely noticed that I was not getting as good service as other people. So I set out to measure. You came in and you waited for your turn; I felt I was not getting a fair deal. I said to myself, `` `Why? No Vice President at IBM said,` Give Hamming a bad time'. It is the secretaries at the

bottom who are doing this. When a slot appears, they'll rush to find someone to slip in, but they go out and find somebody else. Now, why? I haven't mistreated them." Answer, I wasn't dressing the way they felt somebody in that situation should. It came down to just that - I wasn't dressing properly. I had to make the decision - was I going to assert my ego and dress the way I wanted to and have it steadily drain my effort from my professional life, or was I going to appear to conform better? I decided I would make an effort to appear to conform properly. The moment I did, I got much better service. And now, as an old colorful character, I get better service than other people.

> 另一个人格缺陷是自我主张，我将在这个案例中谈谈我自己的经历。我来自洛斯阿拉莫斯，早期我在纽约租用一台计算机。我当时仍然西装革履、大口袋、波波头。我隐约注意到我得到的服务不如其他人好。所以我开始评估这一切。你进来了并排到了你的位次。我觉得我没有得到公平的待遇。为什么。IBM没有任何一个副总裁说：“给Hamming一段糟糕的体验。”是底层的秘书在做这件事情。当一个空位出现时，他们会找到人填补过去。不过为什么，我没有被妥善对待。他们觉得我穿了一个不合时宜的衣服。归根结底，我穿着不得体。我必须做出决定，需要维护自我按照我想要的方式穿着，让它不断耗尽我职业生涯的努力，还是我应该努力符合要求？我决定，我会努力地表现得更加得体。我这样做之后，我得到了更好的服务。现在我甚至得到了比其他人更好的服务。

You should dress according to the expectations of the audience spoken to. If I am going to give an address at the MIT computer center, I dress with a bolo and an old corduroy jacket or something else. I know enough not to let my clothes, my appearance, my manners get in the way of what I care about. An enormous number of scientists feel they must assert their ego and do their thing their way. They have got to be able to do this, that, or the other thing, and they pay a steady price.

> 你应该按照听众的期望来着装。如果我要在麻省理工学院的计算机中心发表演讲，我需要穿一件POLO衫和一个旧灯芯绒夹克或其它什么。我知道不要让我的衣服、外表和举止妨碍我所关心的事情。大量科学家认为他们必须坚持自我，按照自己的方式做事情。他们必须能够做这个、那个或其它事情，并做稳定的付出。

John Tukey almost always dressed very casually. He would go into an important office and it would take a long time before the other fellow realized that this is a first-class man and he had better listen. For a long time John has had to overcome this kind of hostility. It's wasted effort! I didn't say you should conform; I said `The *appearance of`

`conforming* gets you a long way.'' If you chose to assert your ego in any number of ways,` I am going to do it my way," you pay a small steady price throughout the whole of your professional career. And this, over a whole lifetime, adds up to an enormous amount of needless trouble.

> John Tukey通常穿着随意。当他去了一个重要的办公室，过了很长时间，才会有其他人意识到这是一个工作一流的人，他最好认真听。很长一段时间以来，John不得不克服这种敌意，这是白费力气！我不是说你应该服从，我是想说：“合规的外表会让你走 很长的路。”如果你选择以任何方式维护你的自我，“我会按照我的方式来做事情。”那么在你的整个职业生涯中，你都会付出一个小而稳定的代价。而在你的一生中，这些堆积起来就是一大堆不必要的麻烦。

By taking the trouble to tell jokes to the secretaries and being a little friendly, I got superb secretarial help. For instance, one time for some idiot reason all the reproducing services at Murray Hill were tied up. Don't ask me how, but they were. I wanted something done. My secretary called up somebody at Holmdel, hopped the company car, made the hour-long trip down and got it reproduced, and then came back. It was a payoff for the times I had made an effort to cheer her up, tell her jokes and be friendly; it was that little extra work that later paid off for me. By realizing you have to use the system and studying how to get the system to do your work, you learn how to adapt the system to your desires. Or you can fight it steadily, as a small undeclared war, for the whole of your life.

> 通过不厌其烦地给秘书们讲笑话，并表现得很友好，我得到了很多秘书的帮助。例如，有一次，由于某种愚蠢的原因，Murray Hill的复产服务器都被捆绑到一起了。别问我这到底是什么回事，但它们的确捆绑在一起了。我想做点事情。我的秘书打电话给在Holmdel的一个人，跳上公司的车，进行了一个小时的旅行，将其恢复了生产，然后回去了。这就是我努力让秘书振作、讲笑话和友好相处的回报，正是那一点点额外的工作后来对握来说得到了回报。通过意识到你必须使用这个系统并研究如何让这个系统来完成你的工作，你学会了如何通过这个系统实现你的想法。或者，你可以把它当作一场小规模的未经宣布的战争，打一辈子。

And I think John Tukey paid a terrible price needlessly. He was a genius anyhow, but I think it would have been far better, and far simpler, had he been willing to conform a little bit instead of ego asserting. He is going to dress the way he wants all of the time. It applies not only to dress but to a thousand other things; people will continue to fight the system. Not that you shouldn't occasionally!

> 我认为John Tukey无畏地付出了可怕的代价。无论如何，他是一个天才，但是我认为如果他如果再愿意符合要求一些，他能更好，也会做事更加容易。他一直按照自己想要的方式进行穿着，这不仅适用于服装，还适用于其它事物。人们需要继续与系统斗争，但这不是说你偶尔不应该。

When they moved the library from the middle of Murray Hill to the far end, a friend of mine put in a request for a bicycle. Well, the organization was not dumb. They waited awhile and sent back a map of the grounds saying, `Will you please indicate on this map what paths you are going to take so we can get an insurance policy covering you.'' A few more weeks went by. They then asked,` Where are you going to store the bicycle and how will it be locked so we can do so and so." He finally realized that of course he was going to be red-taped to death so he gave in. He rose to be the President of Bell Laboratories.

> 当他们把图书馆从Murray Hill的中间搬到远处时，我的一个朋友提出要一辆自行车。这个组织并不是愚蠢，他们等了一会，寄回了一张场地地图，上面写着：“请你在这张地图上注明你要走什么路，这样我们就可以为你投保了。”又过了几个星期，他们问：“你打算把自行车存储在哪里？怎么把自行车锁上？这样我们可以继续做一些事情。”他终于意识到自己可能会胶带缠死，所以他让步了。他最终升为了贝尔实验室的总裁。

Barney Oliver was a good man. He wrote a letter one time to the IEEE. At that time the official shelf space at Bell Labs was so much and the height of the IEEE Proceedings at that time was larger; and since you couldn't change the size of the official shelf space he wrote this letter to the IEEE Publication person saying, ``Since so many IEEE members were at Bell Labs and since the official space was so high the journal size should be changed." He sent it for his boss's signature. Back came a carbon with his signature, but he still doesn't know whether the original was sent or not. I am not saying you shouldn't make gestures of reform. I am saying that my study of able people is that they don't get themselves committed to that kind of warfare. They play it a little bit and drop it and get on with their work.

> Barney Oliver是一个优秀的，他曾经给IEEE写过一封信。当时，贝尔实验室的官方排版空间很大，而当时IEEE会议给的空间更大。由于你不能改变官方排版的空间大小，他给IEEE出版人员写了这封信，说：“既然有这么多IEEE成员在贝尔实验室，而且官方排版空间太高，期刊的大小应该改变。”他寄来请老板签字，

> 返回的是一个带有他签名的文字，但是他仍然不知道原件是否已经寄出。我并不是说你不应该做出改变的姿态。我的意思是，对于那些有能力的人来，研究应该是不会致力于那种战争。他们进行了一点，放下它，并继续开展自己的工作。

Many a second-rate fellow gets caught up in some little twitting of the system, and carries it through to warfare. He expends his energy in a foolish project. Now you are going to tell me that somebody has to change the system. I agree; somebody's has to. Which do you want to be? The person who changes the system or the person who does first-class science? Which person is it that you want to be? Be clear, when you fight the system and struggle with it, what you are doing, how far to go out of amusement, and how much to waste your effort fighting the system. My advice is to let somebody else do it and you get on with becoming a first-class scientist. Very few of you have the ability to both reform the system and become a first-class scientist.

> 许多二流的人陷入了对于系统的一些小问题中，并将其带入了战争。他把精力花在一个愚蠢的项目中。现在你告诉我，必须有人要改变这个系统。我同意，有人不得不这样做。你想成为哪一个？改变系统的人还是做一流科学的人？你想成为哪个人？要清楚，当你与系统斗争时，你在作什么，有多大程度失去了乐趣，以及在与系统斗争时浪费了多少精力。我的建议是让别人来做，你就可以成为一流的科学家。你们中很少有人能够改变系统并同时成为伟大的科学家。

On the other hand, we can't always give in. There are times when a certain amount of rebellion is sensible. I have observed almost all scientists enjoy a certain amount of twitting the system for the sheer love of it. What it comes down to basically is that you cannot be original in one area without having originality in others. Originality is being different. You can't be an original scientist without having some other original characteristics. But many a scientist has let his quirks in other places make him pay a far higher price than is necessary for the ego satisfaction he or she gets. I'm not against all ego assertion; I'm against some.

> 另一方面，我们不能总是屈服。有时，一定程度上的反抗是明智的。我观察到，几乎所有的科学家都喜欢对这个系统进行一定程度的研究，因为他们非常喜欢它。归根结底，你不可能在一个领域具有独创性，而在另一个领域中没有独创性，这样你就不能成为一名独创的科学家。但是很多科学家让他在其它地方的怪癖使得他付出了比他获得自我满足所需要的代价高得多的代价。我不反对一切的自我主张，我只是反对一部分。

Another fault is anger. Often a scientist becomes angry, and this is no way to handle things. Amusement, yes, anger, no. Anger is misdirected. You should follow and cooperate rather than struggle against the system all the time.

> 另一个错误是愤怒。科学家经常会生气，而这不是处理事情的方法。娱乐是，愤怒不是。愤怒是表明个人被误导了。你应该遵循和合作，而不是一直和系统对抗。

Another thing you should look for is the positive side of things instead of the negative. I have already given you several examples, and there are many, many more; how, given the situation, by changing the way I looked at it, I converted what was apparently a defect to an asset. I'll give you another example. I am an egotistical person; there is no doubt about it. I knew that most people who took a sabbatical to write a book, didn't finish it on time. So before I left, I told all my friends that when I come back, that book was going to be done! Yes, I would have it done - I'd have been ashamed to come back without it! I used my ego to make myself behave the way I wanted to. I bragged about something so I'd have to perform. I found out many times, like a cornered rat in a real trap, I was surprisingly capable. I have found that it paid to say, ``Oh yes, I'll get the answer for you Tuesday," not having any idea how to do it. By Sunday night I was really hard thinking on how I was going to deliver by Tuesday. I often put my pride on the line and sometimes I failed, but as I said, like a cornered rat I'm surprised how often I did a good job. I think you need to learn to use yourself. I think you need to know how to convert a situation from one view to another which would increase the chance of success.

> 你应该寻找的事情积极的一面，而不是消极的一面。我已经给你举了几个例子，而且还有很多很多；在这种情况下，我如何通过改变看待它的方式，把明显的缺陷转化为资产。我再举一个例子。我是一个自负的人，这毫无疑问。我知道大多数休假写书的人都没有按时完成，所以在我离开的时候，我会告诉我的朋友们当我回来的时候，那本书就要写出来了。是的，我会这么做，这样如果那本书没有写完，我会感到羞愧的。我用我的自我让自己按照我想要的方式行事。我吹嘘一些事情，所以我必须付诸行动。我发现很多次，我就像一只被困在真正的陷阱中走投无路的老鼠，我很有能力。我发现，说“我会在下周二给你答案”是值得的，但我其实并不知道应该怎么做。到了周日晚上，我废了很大力气准备周二要回答的内容。我经常把我的骄傲放置在危险当中，有时我会失败，正如我所说的，就像一只走投无路的老鼠一样，我很惊讶自己经常做得很好。我认为你需要学会利

> 用自己，需要了解如何把一件事情的一个样子转变成另一个样子，这样可以增加成功的机会。

Now self-delusion in humans is very, very common. There are enumerable ways of you changing a thing and kidding yourself and making it look some other way. When you ask, `Why didn't you do such and such,'' the person has a thousand alibis. If you look at the history of science, usually these days there are 10 people right there ready, and we pay off for the person who is there first. The other nine fellows say,` Well, I had the idea but I didn't do it and so on and so on." There are so many alibis. Why weren't you first? Why didn't you do it right? Don't try an alibi. Don't try and kid yourself. You can tell other people all the alibis you want. I don't mind. But to yourself try to be honest.

> 人类的自欺欺人现象非常普遍，有无数种方法可以让你改变一件事情，让它看起来与众不同。当你问："为什么不做那样的事情？"时，那个人总是有一千个理由。如果你看看科学史，通常有10个人准备好了，我们会为第一个到达终点的人喝彩。其他九个人说："好吧，我有这个想法，但我没有做到。"他们总是有很多借口和理由。为什么你不是第一个？你为什么做得不对？不要试图找借口。不要自欺欺人。你可以告诉他人你所有的借口，我不介意，但是最好自己对自己尽量诚实。

If you really want to be a first-class scientist you need to know yourself, your weaknesses, your strengths, and your bad faults, like my egotism. How can you convert a fault to an asset? How can you convert a situation where you haven't got enough manpower to move into a direction when that's exactly what you need to do? I say again that I have seen, as I studied the history, the successful scientist changed the viewpoint and what was a defect became an asset.

> 如果你真的想成为一名一流的科学家，你需要了解自己，了解自己的弱点、长处以及一些缺点，比如我的利己主义。如何把缺陷转换成自己的优势？当你没有足够的人力朝着一个方向前进时，你如何才能将这种情况转变为你需要做的事情？我再说一遍，在我研究历史的过程中，我看到了成功的科学见改变了自己的观点，他把缺陷变成了自己的资产。

In summary, I claim that some of the reasons why so many people who have greatness within their grasp don't succeed are: they don't work on important problems, they don't become emotionally involved, they don't try and change what is difficult to some other situation which is easily done but is still important, and they keep giving themselves alibis why they don't. They keep saying that it is a matter of luck. I've told you how easy it is; furthermore I've told you how to reform. Therefore, go forth and become great scientists!

> 总之，我认为许多没有做出伟大工作的人的一些原因是：他们没有解决重要的问题，他们没有付诸感情，他们没有将困难的事情转变为其它容易做到但也十分重要的事情。他们不断为自己的不成功提供借口。他们一直说这是运气的问题，我已经告诉你这是多么容易。此外，我已经告诉你如何改变。因此，走出去，去成为伟大的科学家吧。

(End of the formal part of the talk.)

DISCUSSION - QUESTIONS AND ANSWERS

A. G. Chynoweth: Well that was 50 minutes of concentrated wisdom and observations accumulated over a fantastic career; I lost track of all the observations that were striking home. Some of them are very very timely. One was the plea for more computer capacity; I was hearing nothing but that this morning from several people, over and over again. So that was right on the mark today even though here we are 20 - 30 years after when you were making similar remarks, Dick. I can think of all sorts of lessons that all of us can draw from your talk. And for one, as I walk around the halls in the future I hope I won't see as many closed doors in Bellcore. That was one observation I thought was very intriguing.

> A.G. Chunoweh: 这是在一个梦幻般的职业生涯中累积的50分钟的集中智慧和观察，我已经记不清所有引人注目的观察结果了。其中一些是非常及时的，一个是呼吁增加计算机的容量。今早我就听到几个人在一遍又一遍说。因此，这在今天是正确的。尽管在你发表类似言论的二三十年后的今天，Dick。我能想到我们所有人都能在你的演讲中汲取各种教训。首先，当我将来在大厅中走来走去的时候，我希望贝尔实验室不会有那么多紧闭的门。这是一个我认为非常有趣的观察结果。

Thank you very, very much indeed Dick; that was a wonderful recollection. I'll now open it up for questions. I'm sure there are many people who would like to take up on some of the points that Dick was making.

> 非常非常感谢你，Dick。那是一个美妙的回忆。现在大家可以对其进行提问。我相信很多人想谈谈Dick提出的一些观点。

Hamming: First let me respond to Alan Chynoweth about computing. I had computing in research and for 10 years I kept telling my management, `Get that !&@#% machine out of research. We are being forced to run problems all the time. We can't do research because were too busy operating and running the computing machines.'' Finally the message got through. They were going to move computing out of research to someplace else. I was persona non grata to say the least and I was surprised that people didn't kick my shins because everybody was having their toy taken away from them. I went in to Ed David's office and said,` Look Ed, you've got to give your researchers a machine. If you give them a great big machine, we'll be back in the same trouble we were before, so busy keeping it going we can't think. Give them the smallest machine you can because they are very able people. They will learn how to do things on a small machine instead of mass computing." As far as I'm concerned, that's how UNIX arose. We gave them a moderately small machine and they decided to make it do great things. They had to come up with a system to do it on. It is called UNIX!

> Hamming: 首先，让我回答Alan Chynoweth关于计算的问题。我从事计算机研究，10年来我一直告诉我的管理层，“让机器停止研究。我们总是被迫遇到问题。我们不能做研究，因为我们忙于运行计算机。”最终信息终于发出去了。他们打算把研究中的计算资源挪动到其它地方，至少可以说我不是很受欢迎。我很惊讶人们没有踢我的小腿，因为他们自己的东西没有。我走进Ed David的办公室说：“Ed，你需要给你的研究人员一台机器。如果你给他们一台很大的机器，我们就会回到之前的困境，一直忙着让它正常运行，我们想都不敢想。给他们一台你能给的最小的机器，因为他们都是非常能干的人。他们能够学习如何在小型机器上而不是大规模计算机中做事。据我所知，UNIX就是这样诞生的。我们给了他们一台相当小的机器，他们决定让它做很多事情。他们必须想出一个系统来做这件事情，它被称为UNIX。”

A. G. Chynoweth: I just have to pick up on that one. In our present environment, Dick, while we wrestle with some of the red tape attributed to, or required by, the regulators, there is one quote that one exasperated AVP came up with and I've used it over and over again. He growled that, ``UNIX was never a deliverable!"

> A. G. Chynoweth: 我需要学习一下，Dick，在我们目前的环境中，当我们与监管部门的一些繁文缛节作斗争的时候，有一句话是一位愤怒的AVP提出的，我已经用了很多次了。他咆哮道："UNIX从来都不是可交付的！"

Question: What about personal stress? Does that seem to make a difference?

> Question: 个人压力呢？它会有什么影响吗？

Hamming: Yes, it does. If you don't get emotionally involved, it doesn't. I had incipient ulcers most of the years that I was at Bell Labs. I have since gone off to the Naval Postgraduate School and laid back somewhat, and now my health is much better. But if you want to be a great scientist you're going to have to put up with stress. You can lead a nice life; you can be a nice guy or you can be a great scientist. But nice guys end last, is what Leo Durocher said. If you want to lead a nice happy life with a lot of recreation and everything else, you'll lead a nice life.

> Hamming: 是的，确实如此。如果你不投入自己的情感，那就做不到。在贝尔实验室的大部分时间我都有溃疡的现象。后来我去了海军研究生院后就好多了。但是，如果你想成为一名伟大的科学家，就必须要忍住压力。你可以过上美好的生活，你也可以是一个优秀的人，也可以是一位伟大的科学家。但是优秀的人总是最后停下来，这也是Leo Durocher所说的。如果你想过上美好幸福的生活，有很多娱乐和其它一切，你会过上美好的生活。

Question: The remarks about having courage, no one could argue with; but those of us who have gray hairs or who are well established don't have to worry too much. But what I sense among the young people these days is a real concern over the risk taking in a highly competitive environment. Do you have any words of wisdom on this?

> Question: 关于勇气的言论，没有人反驳。但是我们这些头发花白的人，或者那些地位稳固的人不需要过于担心。但是最近，我在年轻人中感受到了在竞争激烈的环境中承担风险的真正担心。你对此有什么高见吗？

Hamming: I'll quote Ed David more. Ed David was concerned about the general loss of nerve in our society. It does seem to me that we've gone through various periods. Coming out of the war, coming out of Los Alamos where we built the bomb, coming out of building the radars and so on, there came into the mathematics department, and the research area, a group of people with a lot of guts. They've just seen things done; they've just won a war which was fantastic. We had reasons for having courage and therefore we did a great deal. I can't arrange that situation to do it again. I cannot blame the present generation for not having it, but I agree with what you say; I just cannot attach blame to it. It doesn't seem to me they have the desire for greatness; they lack the courage to do it. But we had, because we were in a favorable circumstance to have it; we just came through a tremendously successful war. In the war we were looking very, very bad for a long while; it was a very desperate struggle as you well know. And our success, I think, gave us courage and self confidence; that's why you see, beginning in the late forties through the fifties, a tremendous productivity at the labs which was stimulated from the earlier times. Because many of us were earlier forced to learn other things - we were forced to learn the things we didn't want to learn, we were forced to have an open door - and then we could exploit those things we learned. It is true, and I can't do anything about it; I cannot blame the present generation either. It's just a fact.

> Hamming: 我会引用Ed David的话。他对我们的社会普遍丧失勇气感到担忧。在我看来，我们确实经历了不同的时期。从战争中走出来，从制造炸弹的洛斯阿拉莫斯出来，从制造雷达中出来，来到了数学学院和研究领域，这里有一群有勇气的人。他们只看到了事情完成了，刚刚赢得了一场精彩的战争。我们拥有勇气，因此做到了很多。我无法再让那件事情的重现。我不能责怪现代人没有勇气，但我同意你所说的。在我看来，年轻人没有追求伟大的欲望，他们缺乏勇气做事情。但是我们有，因为我们处于有利的环境中，我们刚刚经历了一场极其成功的战争。在战争中，我们在很长一段时间中看起来非常糟糕。正如你所知，这是一场非常绝望的斗争。我认为，我们的成功给了我们勇气和自信，这就是你为什么能看到，从四十年代末到五十年代，实验室的生产力得到了刺激。因为我们中许多人早些时候被迫学习其他东西，我们被迫学习了不想学习的事情，我们被迫变得开放，然后我们可以利用我们学到的东西。这是真的，我对此无能为力，我也不能责怪当代人。

Question: Is there something management could or should do?

> Question: 管理层可以或者应该做些什么吗？

Hamming: Management can do very little. If you want to talk about managing research, that's a totally different talk. I'd take another hour doing that. This talk is about how the individual gets very successful research done in spite of anything the management does or in spite of any other opposition. And how do you do it? Just as I observe people doing it. It's just that simple and that hard!

> Hamming: 管理层几乎无能为力。如果你想谈论管理研究，那就完全不同了。我会再花一个小时做那件事情。这是关于个人如何在管理层做任何事情或不顾任何其它反对意见的情况下完成非常成功的研究。你是怎么做到的？就像我观察到人们在做这件事情一样，或简单或困难。

Question: Is brainstorming a daily process?

> Question: 每天都进行头脑风暴吗？

Hamming: Once that was a very popular thing, but it seems not to have paid off. For myself I find it desirable to talk to other people; but a session of brainstorming is seldom worthwhile. I do go in to strictly talk to somebody and say, `Look, I think there has to be something here. Here's what I think I see ...'' and then begin talking back and forth. But you want to pick capable people. To use another analogy, you know the idea called the `critical mass.' If you have enough stuff you have critical mass. There is also the idea I used to call `sound absorbers'. When you get too many sound absorbers, you give out an idea and they merely say,` Yes, yes, yes." What you want to do is get that critical mass in action; `Yes, that reminds me of so and so,'' or,` Have you thought about that or this?" When you talk to other people, you want to get rid of those sound absorbers who are nice people but merely say, ``Oh yes," and to find those who will stimulate you right back.

> Hamming: 曾经这是一件很受欢迎的事情，但是似乎没有得到回报。我的确会认真地和他人交谈，但是集思广益可能是不值得的。我会说：“看，我想这里一定有什么科学道路，这是我想看到的...” 接着会进行多次交流。但是你想要选择有能力的人。使用另一个比喻，你知道所谓的临界质量。如果你有足够的东西，就会存在临界质量。还有一个我过去称之为吸声器的想法。当你有太多的吸声器的时候，你发出一个想法他们只会说对。你想做的是让临界质量发挥作用：“是的，这让我想起了某某”或“你想过这个或那个吗？”当你和其他人交谈时，你想

> 摆脱那些吸声器，他们是优秀的人，但是只会说是的，你需要找到那些会给你反馈的人。

For example, you couldn't talk to John Pierce without being stimulated very quickly. There were a group of other people I used to talk with. For example there was Ed Gilbert; I used to go down to his office regularly and ask him questions and listen and come back stimulated. I picked my people carefully with whom I did or whom I didn't brainstorm because the sound absorbers are a curse. They are just nice guys; they fill the whole space and they contribute nothing except they absorb ideas and the new ideas just die away instead of echoing on. Yes, I find it necessary to talk to people. I think people with closed doors fail to do this so they fail to get their ideas sharpened, such as ``Did you ever notice something over here?" I never knew anything about it - I can go over and look. Somebody points the way. On my visit here, I have already found several books that I must read when I get home. I talk to people and ask questions when I think they can answer me and give me clues that I do not know about. I go out and look!

> 例如，你不能在没有很快反馈的情况喜爱与John Pierce交谈。我曾经和一群其他人交谈，例如Ed Gilbert，我过去经常去他的办公室问他问题，然后兴奋地回来。吸声器也是优秀的人，但是他们只是吸收思想却没有反馈。他们占据了整个空间，却什么也没做。是的，我觉得有必要与人们交谈。我认为那些关起门来的人没有做到这一点，所以他们 的想法也没有变得尖锐，比如“你注意到这里有什么吗？”我对此一无所知，但是我可以过去看看。在我看到这里时，我已经找到了几本我回家之后必须阅读的书籍。当我认为人们能回答我并给我一些不知道的线索时，我会与他们交谈并提出问题。

Question: What kind of tradeoffs did you make in allocating your time for reading and writing and actually doing research?

> Question: 你在分配阅读、写作和实际研究的时间方面做了什么权衡？

Hamming: I believed, in my early days, that you should spend at least as much time in the polish and presentation as you did in the original research. Now at least 50% of the time must go for the presentation. It's a big, big number.

> Hamming: 在早期的时候，我认为你应该至少花最初研究一样多的时间用来打磨和展示。现在至少50%的时间必须用于展示，这是一个很大的数字。

Question: How much effort should go into library work?

> Question: 应该在图书馆中投入多少精力？

Hamming: It depends upon the field. I will say this about it. There was a fellow at Bell Labs, a very, very, smart guy. He was always in the library; he read everything. If you wanted references, you went to him and he gave you all kinds of references. But in the middle of forming these theories, I formed a proposition: there would be no effect named after him in the long run. He is now retired from Bell Labs and is an Adjunct Professor. He was very valuable; I'm not questioning that. He wrote some very good Physical Review articles; but there's no effect named after him because he read too much. If you read all the time what other people have done you will think the way they thought. If you want to think new thoughts that are different, then do what a lot of creative people do - get the problem reasonably clear and then refuse to look at any answers until you've thought the problem through carefully how you would do it, how you could slightly change the problem to be the correct one. So yes, you need to keep up. You need to keep up more to find out what the problems are than to read to find the solutions. The reading is necessary to know what is going on and what is possible. But reading to get the solutions does not seem to be the way to do great research. So I'll give you two answers. You read; but it is not the amount, it is the way you read that counts.

> Hamming: 这取决于研究的领域。贝尔实验室中有一个非常聪明的人，他经常待在图书馆中，什么都读。如果你想得到一些引用，你可以去找他，他会给你一堆各种各样的引用。但是形成这些理论之前，我得到了一个主张，从长远角度看，不会有以他的名字命名的那种影响力。他现在已经从贝尔实验室退休了，是一名兼职教授。他非常有价值，我从未质疑这一点。他写了很多好的《物理评论》的文章， 但是没有足够的影响力，因为他阅读太多了。如果你一直阅读别人做过的东西，你就会按照他们思考的那样去做。如果你想要得到一些不同的新想法，就应该做大多数有创造力的人做的事情，把问题弄得非常清楚，拒绝看任何答案，直到你仔细思考了这个问题，你应该如何去做，你应该如何略微修改一下这个问题，使其变得正确。所以，是的，你需要一直紧跟脚步。你需要更多地了解问题所在，而不是通过阅读来找到解决方案。阅读是必要的，以了解正在发生的事情以及可能发生的事情。但是阅读以获得解决方案似乎不是进行伟大研究的方法。所以我会给你两个答案，你应该阅读，但是真正重要的不是阅读的总量，而是你阅读的方式。

Question: How do you get your name attached to things?

> Question: 你如何让你的名字与事物联系起来？

Hamming: By doing great work. I'll tell you the hamming window one. I had given Tukey a hard time, quite a few times, and I got a phone call from him from Princeton to me at Murray Hill. I knew that he was writing up power spectra and he asked me if I would mind if he called a certain window a `Hamming window.'' And I said to him,` Come on, John; you know perfectly well I did only a small part of the work but you also did a lot." He said, ``Yes, Hamming, but you contributed a lot of small things; you're entitled to some credit." So he called it the hamming window. Now, let me go on. I had twitted John frequently about true greatness. I said true greatness is when your name is like ampere, watt, and fourier - when it's spelled with a lower case letter. That's how the hamming window came about.

> Hamming: 通过做伟大的工作。例如Hamming Window One，我曾经多次让Tukey难过，我在Murray Hill接到了他从Princeton的电话。我知道他在写功率谱，他问我是否介意他把某个窗口成为Hamming Window。我说可以的John，你很清楚我只做了一小部分工作，但是你做了很多。他说，是的，Hamming，但是你贡献了很多小的东西，你有资格获得一些荣誉并把它称为Hamming Window。现在我让继续说下去，我经常在真正的伟大方面挖苦John。我说真正的伟大是你的名称像Ampere、Watt或者Fourier，当它是通过小写字母拼写时，这也就是hamming window的来源了。

Question: Dick, would you care to comment on the relative effectiveness between giving talks, writing papers, and writing books?

> Question: Dick，你能谈谈演讲、写论文和写书之间的相互影响吗？

Hamming: In the short-haul, papers are very important if you want to stimulate someone tomorrow. If you want to get recognition long-haul, it seems to me writing books is more contribution because most of us need orientation. In this day of practically infinite knowledge, we need orientation to find our way. Let me tell you what infinite knowledge is. Since from the time of Newton to now, we have come close to doubling knowledge every 17 years, more or less. And we cope with that, essentially, by specialization. In the next 340 years at that rate, there will be 20 doublings, i.e. a million, and there will be a million fields of specialty for every one field now. It isn't going to happen. The present growth of knowledge will choke itself off until we get different tools. I believe that books which try to digest, coordinate, get rid of the duplication, get rid of the less fruitful

methods and present the underlying ideas clearly of what we know now, will be the things the future generations will value. Public talks are necessary; private talks are necessary; written papers are necessary. But I am inclined to believe that, in the long-haul, books which leave out what's not essential are more important than books which tell you everything because you don't want to know everything. I don't want to know that much about penguins is the usual reply. You just want to know the essence.

> Hamming: 从短期来看，如果你想明天就刺激某个人，论文是非常重要的。如果你想长期受到认可，在我看来写书是更大的共线，因为我们大多数人需要指引。在这个似乎知识无限的时代，我们需要指引来找到我们自己的路。让我告诉你什么是无限的知识，从牛顿时代到现在，我们几乎每17年就会增加一倍的知识。我们主要通过专业化的方式应对这个问题。在接下来的340年中，按照这个速度，将会增长2^20次方个，即百万级，这样每个领域就会有100万个专业领域。这应该不会发生。在我们获取不同的工具之前，目前的知识增长会将自我扼杀。我相信，那些试图消化、协调、消除重复、消除不太富有成效的方法并将基本思想清晰地呈现在我们面前的书籍，将是子孙后代所珍视的东西。公开的演讲是必要的，私下的交流也是必要的，写论文是必要的。我倾向于相信，从长远看，那些忽略了非必要内容的书籍比因为你不想知道一切而告诉你一切的书更加重要。我不想知道那么多无关的事情，你只想知道本质。

Question: You mentioned the problem of the Nobel Prize and the subsequent notoriety of what was done to some of the careers. Isn't that kind of a much more broad problem of fame? What can one do?

> Question: 你提到了诺贝尔奖的问题，以及对随后职业生涯的一些不好的影响？这不是一个更加广泛的名声问题吗？那个人可以做到什么？

Hamming: Some things you could do are the following. Somewhere around every seven years make a significant, if not complete, shift in your field. Thus, I shifted from numerical analysis, to hardware, to software, and so on, periodically, because you tend to use up your ideas. When you go to a new field, you have to start over as a baby. You are no longer the big mukity muk and you can start back there and you can start planting those acorns which will become the giant oaks. Shannon, I believe, ruined himself. In fact when he left Bell Labs, I said, `That's the end of Shannon's scientific career.'' I received a lot of flak from my friends who said that Shannon was just as smart as ever. I said,` Yes, he'll be just as smart, but that's the end of his scientific career," and I truly believe it was.

> Hamming: 你可以做的事情如下。大约每7年你所在的领域会发生一次大的转变。因此，我周期性地转向数值分析、硬件、软件等，因为你往往会耗尽你的想法。当你进入一个新的领域的时候，你就需要从婴儿时期开始，你不再是那个参天大树。但是你可以开始种植橡子，并期待它长成巨大的橡树。我相信Shannon毁掉了自己，事实上，当他离开贝尔实验室的时候，我说这就是Shannon科学生涯的结束。我收到了很多朋友的指责，他们说Shannon和之前一样聪明，我说："是的，他也会同样聪明，但这就是他科学生涯的结束。"我真的相信就是这样。

You have to change. You get tired after a while; you use up your originality in one field. You need to get something nearby. I'm not saying that you shift from music to theoretical physics to English literature; I mean within your field you should shift areas so that you don't go stale. You couldn't get away with forcing a change every seven years, but if you could, I would require a condition for doing research, being that you will change your field of research every seven years with a reasonable definition of what it means, or at the end of 10 years, management has the right to compel you to change. I would insist on a change because I'm serious. What happens to the old fellows is that they get a technique going; they keep on using it. They were marching in that direction which was right then, but the world changes. There's the new direction; but the old fellows are still marching in their former direction.

> 你必须改变，过了一段时间你就会累，你会在一个领域耗尽你的独创性。你需要在附近领域找点相关的内容做。我并不是说你应该从音乐转向理论物理学，再转向英文文学。我的意思是，在你的领域中，你应该改变领域，这样你就不会过时。你不能强迫自己每7年改变一次，如果可以的话，我会要求你有一个做研究的条件，那就是每7年改变一次你的研究领域，并且对其含义能够有一个合理的定义，或者在10年后，管理层有权迫使你去改变。我会坚持改变，因为我是认真的。老家伙们的遭遇是，他们掌握了一种技巧，他们一直在使用它。他们朝着当时正确的方向前进，但是世界发生了变化，有的新的方向，但是这些老家伙仍然朝着他们原来的方向前进。

You need to get into a new field to get new viewpoints, and before you use up all the old ones. You can do something about this, but it takes effort and energy. It takes courage to say, `Yes, I will give up my great reputation.'' For example, when error correcting codes were well launched, having these theories, I said,` Hamming, you are going to quit reading papers in the field; you are going to ignore it completely; you are going to try and do something else other than coast on that."

I deliberately refused to go on in that field. I wouldn't even read papers to try to force myself to have a chance to do something else. I managed myself, which is what I'm preaching in this whole talk. Knowing many of my own faults, I manage myself. I have a lot of faults, so I've got a lot of problems, i.e. a lot of possibilities of management.

> 你需要进入一个新的领域来获取新的观点，最好在你用完所有的旧的想法之前。你可以为此做点什么，但是这需要付出努力和精力，说："是的，我需要放弃我的名誉。"这也需要勇气，当纠错码被提出之后，有了这些理论后，我说："Hamming，你需要提供阅读这个领域的文章，你需要彻底忽略它们，你应该做点其他的事情。"我故意拒绝再在那个领域继续工作，我甚至不会再阅读报纸来强迫自己有机会做那件事情。我管理好了自己，这就是我在整个演讲过程中所宣讲的。我知道自己的很多缺点，所以我能控制住自己。我有很多缺点，有很多问题，管理的可能性也就很大。

Question: Would you compare research and management?

> Question: 你会比较研究和管理吗？

Hamming: If you want to be a great researcher, you won't make it being president of the company. If you want to be president of the company, that's another thing. I'm not against being president of the company. I just don't want to be. I think Ian Ross does a good job as President of Bell Labs. I'm not against it; but you have to be clear on what you want. Furthermore, when you're young, you may have picked wanting to be a great scientist, but as you live longer, you may change your mind. For instance, I went to my boss, Bode, one day and said, `Why did you ever become department head? Why didn't you just be a good scientist?'' He said,` Hamming, I had a vision of what mathematics should be in Bell Laboratories. And I saw if that vision was going to be realized, I had to make it happen; I had to be department head." When your vision of what you want to do is what you can do single-handedly, then you should pursue it. The day your vision, what you think needs to be done, is bigger than what you can do single-handedly, then you have to move toward management. And the bigger the vision is, the farther in management you have to go. If you have a vision of what the whole laboratory should be, or the whole Bell System, you have to get there to make it happen. You can't make it happen from the bottom very easily. It depends upon what goals and what desires you have. And as they change in life, you have to be prepared to change. I chose to avoid management because I preferred to do what I could do single-handedly. But that's the choice that I made, and it is biased. Each person is entitled to their choice. Keep an open

mind. But when you do choose a path, for heaven's sake be aware of what you have done and the choice you have made. Don't try to do both sides.

> Hamming: 如果你想成为一名优秀的研究人员，你就无法胜任公司总裁的工作。如果你想当公司的总裁，那就是另一回事。我并不反对当这家公司的总裁，我只是不想成为。我认为Ian Ross总裁在贝尔实验室做得很好。我并不反对这样。但你必须清楚自己想要什么。此外，当你年轻的时候，你可能选择了成为一名伟大的科学家，但随着你年长，你可能会改变主意。例如，有一天，我去找我的老板Bode，问："你为什么要当部门主管？你为什么不做一个好的科学家？"他说："Hamming，我对贝尔实验室的数学应该发展成什么有一个愿景。如果这个愿景实现，就只能是通过我来当部门主管的方式。"当你对自己想做的事情的愿景是你一个人能做的，你就应该追求它。当你的愿景，你认为需要做的事情，比你一个人能够做的事情更大的时候，你就需要走向管理层。愿景越大，你在管理方面就必须走得更远。如果你对整个实验室或者整个贝尔系统有一个愿景，你就必须要去做总裁。你不可能很容易地从底层实现它。这取决于你的目标和欲望。当它们在生活中发生变化时，你必须做好改变的准备。我选择避开管理层，因为我更喜欢独自做我能做的事情。但这是我做出的选择，而且是有偏见的。每个人都有自己的选择权，保持开放的心态。但是，当你选择一条路上，要意识到你所做的一切和你所做出的选择。不要试图两全其美。

Question: How important is one's own expectation or how important is it to be in a group or surrounded by people who expect great work from you?

> Question: 一个人自己的期望有多重要，或者在一个团队中或周围人有人期望你做得好有多重要？

Hamming: At Bell Labs everyone expected good work from me - it was a big help. Everybody expects you to do a good job, so you do, if you've got pride. I think it's very valuable to have first-class people around. I sought out the best people. The moment that physics table lost the best people, I left. The moment I saw that the same was true of the chemistry table, I left. I tried to go with people who had great ability so I could learn from them and who would expect great results out of me. By deliberately managing myself, I think I did much better than laissez faire.

> Hamming: 在贝尔实验室中，每个人都希望我能做得好，这是一个很大的帮助。每个人都希望你做得好，如果你有自豪感，你也会这样做。我认为有一流的人才在身边是非常有价值的。我寻找最优秀的人。当那个物理组餐桌失去了最好的人时，我离开了。当我看到化学组餐桌也是这样时，我也离开了。我试着去找那些真正有能力的人，这样我就可以像他们学习，也可以期待我能够取得好成绩。通过刻意管理自己，我认为我做得比自由放纵时好得多。

Question: You, at the outset of your talk, minimized or played down luck; but you seemed also to gloss over the circumstances that got you to Los Alamos, that got you to Chicago, that got you to Bell Laboratories.

> Question: 你在演讲一开始就尽量减少或者淡化运气，但是你似乎也掩盖了把你带到洛斯阿拉莫斯、芝加哥、贝尔实验室的真实情况。

Hamming: There was some luck. On the other hand I don't know the alternate branches. Until you can say that the other branches would not have been equally or more successful, I can't say. Is it luck the particular thing you do? For example, when I met Feynman at Los Alamos, I knew he was going to get a Nobel Prize. I didn't know what for. But I knew darn well he was going to do great work. No matter what directions came up in the future, this man would do great work. And sure enough, he did do great work. It isn't that you only do a little great work at this circumstance and that was luck, there are many opportunities sooner or later. There are a whole pail full of opportunities, of which, if you're in this situation, you seize one and you're great over there instead of over here. There is an element of luck, yes and no. Luck favors a prepared mind; luck favors a prepared person. It is not guaranteed; I don't guarantee success as being absolutely certain. I'd say luck changes the odds, but there is some definite control on the part of the individual.

> Hamming: 运气的确不错。另一方面我也不知道还有可选项。除非你能说在其它机构就不会同样或者更加成功。你能说这件事就是运气好吗？例如，当我在洛斯阿拉莫斯遇到Feynman时，我知道他会获得诺贝尔奖。我清楚他会做得很好，果不其然，他就是做得很好。这并不是说你在这种情况下只做了一丁点伟大的工作，这是运气，你迟早会有很多机会。如果你处于这种情况，你抓住了其中的一个机会，你在那里而不是另一个地方，这可能是运气成分。运气会帮助有准备的头脑，运气偏爱有准备的人。我不保证一定能成功，我想说运气确实会改变成功的几率，但是个人也确实对此有一定控制力。

Go forth, then, and do great work!

> 那就去吧，去做好的工作！

(End of the General Research Colloquium Talk.)